

Experimental Data for Introductory Solid State Physics

Ozan Zeray

*A companion to the topics of introductory solid state physics,
presenting the real experimental data behind the theory.*

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About This Book

I am writing this book to provide experimental demonstrations and data for better understanding of introductory solid state physics subjects.

In my experience, I have found this to be completely omitted by theoretical considerations and arguments. Thus here I gathered the very real experimental data straight from the very real life which shows very real physics effects and responses to various experimentation. This book will be furthermore an aid for the curious who craves the very real phenomena rather than diagrams, graphs or other absurdly strange ways of obstructing the very real results into something completely irrelevant for the sake of 'simplification' or 'education'.

When I heard about the subject 'Solid State Physics' in highschool, I couldn't think of any other study which could be less boring than this. My very first instinct was to think about solids and anything interesting about them. However I could only think of stones, pebbles, rocks, wood, iron, hard plastic and other common items. It was my problem to not have solids of dozens of different elements in the periodic table at my disposal and not have conducted various electrical, thermodynamics or optics experiments with them. Therefore to attract your attention to this matter, I have here gathered the data and experiments which scientists worked on through the years that show very curious patterns and indicate towards greater rules of the constitution of the universe. It is my pleasure to present you this accumulated data on behaviours of solids.

"Ben manevi miras olarak hicbir nas-i kati, hicbir dogma, hicbir donmus, kaliplasmus kural birakmiyorum. Benim manevi mirasim bilim ve akildir. Benden sonra, beni benimsemek isteyenler, bu temel mihver uzerinde aklın ve bilimin rehberligini kabul ederlerse, manevi mirasclarım olurlar."

>

1881 -- infinity

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Part I: Crystal Structure, Diffraction, and Binding

Chapter 1: Crystal Structure

An ideal crystal is constructed by the infinite repetition of identical groups of atoms. A group is called the basis. The set of mathematical points to which the basis is attached is called the lattice. The lattice is a periodic array of points in space; the basis is what sits on each point.

1.1 Periodic Array of Atoms

A lattice is defined by three fundamental translation vectors $\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3$ such that any lattice point $\mathbf{R}$ can be written as:

[equation]

where n_1, n_2, n_3 are integers.

As W. L. Bragg wrote in 1913: "In treating the diffraction of waves by a space point system such as a crystal, that case is the most simple in which the diffraction is caused by a series of points arranged in a space lattice, of one of the 14 Bravais types. Here every point is identical with every other point of the arrangement, and it is always possible to find an element of the pattern consisting of a parallelepiped with a point at each corner." [Bragg 1913, p. 249]

The crystal structure is then given by specifying the lattice and the basis. The same lattice can describe very different crystals if the basis is changed.

1.2 Fundamental Types of Lattices

In two dimensions there are exactly 5 distinct lattice types. In three dimensions, Auguste Bravais showed in 1848 that there are exactly 14 distinct lattice types, classified into 7 crystal systems.

System	Axes	Angles	Bravais Lattices
Cubic	$a = b = c$	$\alpha = \beta = \gamma = 90^\circ$	sc, bcc, fcc
Tetragonal	$a = b \neq c$	$\alpha = \beta = \gamma = 90^\circ$	simple, body-centered
Orthorhombic	$a \neq b \neq c$	$\alpha = \beta = \gamma = 90^\circ$	simple, base, body, face
Hexagonal	$a = b \neq c$	$\alpha = \beta = 90^\circ, \gamma = 120^\circ$	simple
Trigonal	$a = b = c$	$\alpha = \beta = \gamma \neq 90^\circ$	simple
Monoclinic	$a \neq b \neq c$	$\alpha = \gamma = 90^\circ \neq \beta$	simple, base-centered
Triclinic	$a \neq b \neq c$	$\alpha \neq \beta \neq \gamma$	simple

Index System for Crystal Planes (Miller Indices)

A crystal plane is specified by three integers (hkl), the Miller indices. The plane makes intercepts $a/h, b/k, c/l$ on the crystal axes.

Bragg introduced these to X-ray crystallography in his 1913 paper (where he called them "Millerian indices"): "There can be assigned to each set of parallel planes integral indices (h, k, l)... Here the integers h, k, l, are reciprocal to the intercepts which a parallel plane makes on the axis of reference." [Bragg 1913, p. 250]

1.3 Simple Crystal Structures

Experimentally Measured Lattice Parameters

The data in the table below are from X-ray diffraction measurements accumulated over more than a century of crystallography, beginning with Bragg's 1913 determination of the NaCl structure.

Crystal	Structure Type	a (Å)	Source
NaCl	Rock salt (B1)	5.640	X-ray diffraction
KCl	Rock salt (B1)	6.293	X-ray diffraction
LiF	Rock salt (B1)	4.027	X-ray diffraction
MgO	Rock salt (B1)	4.211	X-ray diffraction
CsCl	Cesium chloride (B2)	4.123	X-ray diffraction
CsBr	Cesium chloride (B2)	4.286	X-ray diffraction
C (diamond)	Diamond (A4)	3.567	X-ray diffraction
Si	Diamond (A4)	5.431	X-ray diffraction
Ge	Diamond (A4)	5.658	X-ray diffraction
GaAs	Zinc blende (B3)	5.653	X-ray diffraction
ZnS	Zinc blende (B3)	5.409	X-ray diffraction
InSb	Zinc blende (B3)	6.479	X-ray diffraction
Cu	fcc (A1)	3.615	X-ray diffraction
Al	fcc (A1)	4.050	X-ray diffraction
Au	fcc (A1)	4.078	X-ray diffraction
Fe	bcc (A2)	2.867	X-ray diffraction
W	bcc (A2)	3.165	X-ray diffraction
Na	bcc (A2)	4.225	X-ray diffraction

1.3.1 Sodium Chloride Structure

The sodium chloride structure (Strukturbericht designation B1, space group $Fm\bar{3}m$) consists of two interpenetrating fcc lattices of Na⁺ and Cl⁻ ions, displaced by half a lattice parameter along each axis.

This was the very first crystal structure ever determined. In his 1913 paper, W. L. Bragg analyzed the X-ray diffraction patterns of NaCl, KCl, KBr, and KI, comparing them systematically to deduce the arrangement of atoms [Bragg 1913].

Bragg's Original Argument (1913)

Bragg's reasoning, extracted directly from his paper, proceeded as follows:

1. KCl appears as a simple cubic lattice. The diffraction pattern of KCl shows a complete set of spots for all integer indices -- as expected for a simple cubic lattice. This is because K⁺ and Cl⁻ are isoelectronic (both have 18 electrons), so they scatter X-rays almost identically. As Bragg wrote: "In the case of potassium chloride the atoms of potassium and chlorine, of atomic weight 39 and 35.5 respectively, are sufficiently close in atomic weight to act as identical diffracting centres." [Bragg 1913, p. 264]
2. KBr shows an fcc pattern. In KBr, the atomic weights differ more (K = 39, Br = 80), and the diffraction pattern shows the characteristic fcc selection rule: only reflections with all-odd or all-even (hkl) are present. Bragg's Tables II and III (p. 257)

show this directly -- the KBr pattern has systematic absences that the ZnS pattern does not.

3. NaCl is intermediate. The NaCl pattern (Tables VII-IX, p. 263) at thin crystal sections (1 mm) resembles KCl, but at thicker sections (6 mm) resembles KBr. This is because Na (atomic weight 23) and Cl (atomic weight 35.5) differ enough to produce different scattering amplitudes for the two sublattices.

4. The structure. Bragg deduced that in all alkali halides, the atoms are arranged on two interpenetrating fcc lattices (Fig. 10, p. 265). He provided three rules [Bragg 1913, p. 264]:

- "There are equal numbers of black and white [atoms]" - "The arrangement of points black and white taken all together is that of the first cubic space lattice" [simple cubic] - "The arrangement of blacks alone or of whites alone is that of the third cubic space lattice" [fcc]

5. The lattice constant. From Fig. 10 (p. 265), Bragg gives the nearest-neighbor distance in NaCl as $AB = 2.8 \times 10^{-8}$ cm (= 2.81 Å), corresponding to a cubic lattice parameter $a = 2$ times 2.81 = 5.62 Å.

Historical note: W. L. Bragg was 25 years old when he published this paper. He and his father W. H. Bragg were awarded the 1915 Nobel Prize in Physics -- W. L. Bragg remains the youngest Nobel laureate in physics.

Alkali Halide Lattice Parameters

	F-	Cl-	Br-
Li?	4.027	5.130	5.501
Na?	4.620	5.640	5.977
K?	5.347	6.293	6.600
Rb?	5.630	6.581	6.889

All values in Å, from X-ray diffraction.

1.3.2 Cesium Chloride Structure

The cesium chloride structure (B2, space group $Pm\bar{3}m$) is not body-centered cubic, despite its visual appearance. It consists of two interpenetrating simple cubic lattices. Each Cs^+ ion is surrounded by 8 Cl^- ions at the corners of a cube (coordination number 8), compared to 6 in the NaCl structure.

CsCl, CsBr, and CsI crystallize in this structure at atmospheric pressure, with lattice constants $a = 4.123$ Å, 4.286 Å, and 4.567 Å respectively.

1.3.3 Hexagonal Close-Packed Structure

The hcp structure is not a Bravais lattice; it is a simple hexagonal lattice with a two-atom basis. For ideal close packing, the ratio $c/a = \sqrt{8/3}$ approx 1.633.

Metal	a (Å)	c/a
Be	2.286	1.568
Mg	3.209	1.624
Ti	2.951	1.587
Zn	2.665	1.856
Cd	2.979	1.886
Co	2.507	1.623
Zr	3.232	1.593

He (hcp, 0 K)	3.570	1.633
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The deviations from ideality are significant. Zinc and cadmium have c/a ratios much larger than ideal, indicating weaker bonding between hexagonal layers. Only helium at 0 K and magnesium come close to the ideal ratio.

1.3.4 Diamond Structure

The diamond structure (A4, space group $Fd\bar{3}m$) is an fcc lattice with a two-atom basis: atoms at (0,0,0) and (1/4, 1/4, 1/4). Each atom is tetrahedrally bonded to four nearest neighbors.

Bond lengths $d = a\sqrt{3}/4$:

- * C (diamond): $d = 1.545 \text{ \AA}$
- * Si: $d = 2.352 \text{ \AA}$
- * Ge: $d = 2.450 \text{ \AA}$

1.3.5 Cubic Zinc Sulfide Structure

The zinc blende structure (B3, space group $F\bar{4}3m$) is identical to diamond but with two different atoms. III-V semiconductors (GaAs, InSb, GaP) and II-VI compounds (ZnS, CdTe) crystallize in this structure.

The key difference from diamond: zinc blende lacks inversion symmetry. This has consequences for piezoelectricity (Ch. 16) and nonlinear optics.

Bragg analyzed zincblende in his 1913 paper alongside the alkali halides. His Tables IV and V (p. 257) separate ZnS reflections into odd-index and even-index planes, establishing the fcc selection rule that would become fundamental to all of crystallography.

1.4 Direct Imaging of Atomic Structure

1.4.1 Transmission Electron Microscopy

The first direct images of a crystal lattice were obtained by J. W. Menter in 1956, who used transmission electron microscopy to image the lattice fringes of platinum phthalocyanine crystals with a spacing of $12 \text{ \AA}$ [Menter 1956]. This was a landmark -- for the first time, the periodic arrangement of atoms in a crystal could be seen rather than merely inferred from diffraction patterns.

Modern aberration-corrected TEM can resolve individual atomic columns. Zhang et al. (2018) demonstrated atomic-resolution imaging of electron beam-sensitive crystalline materials using HRTEM, published in Science [Zhang 2018].

1.4.2 Scanning Tunneling Microscopy

The scanning tunneling microscope (STM), invented by Gerd Binnig and Heinrich Rohrer at IBM Zürich in 1981 (Nobel Prize 1986), provides real-space images of surfaces with atomic resolution. The STM measures the quantum mechanical tunneling current between a sharp metallic tip and the sample surface.

The iconic STM image of solid state physics is the Si(111)- 7×7 surface reconstruction, which shows the arrangement of 49 atoms in the surface unit cell -- a structure that could not have been determined by diffraction alone.

In 1990, Don Eigler and Erhard Schweizer at IBM Almaden positioned 35 xenon atoms on a nickel surface to spell "IBM" -- the first demonstration of atomic-scale engineering [Eigler 1990]. We discuss this in detail in Chapter 18.

1.5 Crystal Structure Data

Ionic Crystal Radii

R. D. Shannon compiled the most widely used set of effective ionic radii in 1976, based on a critical evaluation of over 1000 interatomic distances from X-ray and neutron diffraction studies.

Ion	Radius (Å)	Ion	Radius (Å)
Li ⁺	0.76	F ⁻	1.33
Na ⁺	1.02	Cl ⁻	1.81
K ⁺	1.38	Br ⁻	1.96
Rb ⁺	1.52	I ⁻	2.20
Cs ⁺	1.67	O ²⁻	1.40
Mg ²⁺	0.72	S ²⁻	1.84
Ca ²⁺	1.00	Se ²⁻	1.98
Sr ²⁺	1.18	Te ²⁻	2.21
Ba ²⁺	1.35		

Coordination number VI (octahedral). From Shannon (1976).

Radius Ratio Rules

The ratio r_+/r_- governs which structure type is stable:

Radius ratio	Structure	Coordination
$r_+/r_- > 0.732$	CsCl	8
$0.414 < r_+/r_- < 0.732$	NaCl	6
$0.225 < r_+/r_- < 0.414$	Zinc blende	4

For NaCl: $r_{\text{Na}^+}/r_{\text{Cl}^-} = 1.02/1.81 = 0.56$, which falls in the NaCl stability range. For CsCl: $r_{\text{Cs}^+}/r_{\text{Cl}^-} = 1.67/1.81 = 0.92$, correctly predicting the CsCl structure.

References

1. Bragg, W. L. (1913). "The Structure of Some Crystals as Indicated by Their Diffraction of X-rays." Proc. R. Soc. Lond. A 89, 248-277. DOI: 10.1098/rspa.1913.0083. -- The paper that founded X-ray crystallography. Contains the first determination of the NaCl structure, the fcc selection rule, and diffraction data for ZnS, KCl, KBr, KI, NaCl.
2. Menter, J. W. (1956). "The Direct Study by Electron Microscopy of Crystal Lattices and Their Imperfections." Proc. R. Soc. Lond. A 236, 119-135. DOI: 10.1098/rspa.1956.0117. -- First direct TEM images of crystal lattice fringes.
3. Zhang, D. et al. (2018). "Atomic-Resolution Transmission Electron Microscopy of Electron Beam-Sensitive Crystalline Materials." Science 359, 675-679. DOI: 10.1126/science.aao0865.
4. Eigler, D. M. and Schweizer, E. K. (1990). "Positioning Single Atoms with a Scanning Tunnelling Microscope." Nature 344, 524-526. DOI: 10.1038/344524a0. -- 35 Xe atoms positioned on Ni(110) to spell "IBM." 2708 citations.
5. Shannon, R. D. (1976). "Revised Effective Ionic Radii and Systematic Studies of Interatomic Distances in Halides and Chalcogenides." Acta Crystallogr. A 32, 751-767.

Chapter 2: Wave Diffraction and the Reciprocal Lattice

The determination of crystal structures is an experimental science. The primary tool is the diffraction of waves -- X-rays, neutrons, or electrons -- by the periodic arrangement of atoms in a crystal. In this chapter we present the experimental foundations: the Bragg law, the reciprocal lattice, the structure factor, and the landmark diffraction experiments that established our understanding of crystal structure and wave-particle duality.

2.1 Diffraction of Waves by Crystals

2.1.1 Bragg Law

In 1912, Max von Laue suggested that X-rays, with wavelengths comparable to interatomic spacings ($\sim 1 \text{ \AA}$), should be diffracted by crystals. The experiment was performed by Friedrich and Knipping, producing the first X-ray diffraction pattern from a crystal of copper sulfate.

The following year, W. L. Bragg provided the simple geometric interpretation [Bragg 1913]. Consider a set of parallel lattice planes with spacing d . An X-ray beam incident at angle θ to the planes is reflected specularly from each plane. Constructive interference occurs when:

[equation]

This is the Bragg law -- the single most important equation in crystallography.

Bragg's Original Verification (1913)

Bragg verified this law by analyzing diffraction from alkali halides. His key experimental observations (from Tables II-IX of the original paper):

- * KCl: All reflections present $\rightarrow$ appears as simple cubic (because K^+ and Cl^- are isoelectronic with 18 electrons each)
- * KBr: Only reflections with all-odd or all-even (hkl) $\rightarrow$ fcc selection rule (because $\text{K} = 39$, $\text{Br} = 80$ differ strongly in scattering power)
- * NaCl: Intermediate behavior $\rightarrow$ both sublattices visible but with different intensities ($\text{Na} = 23$, $\text{Cl} = 35.5$)

"In the case of potassium chloride the atoms of potassium and chlorine, of atomic weight 39 and 35.5 respectively, are sufficiently close in atomic weight to act as identical diffracting centres." -- Bragg 1913, p. 264

2.1.2 The Debye-Scherrer Powder Method

In 1916, Debye and Scherrer demonstrated that crystalline powders also produce useful diffraction patterns. Because a powder contains crystallites in all orientations, each set of planes (hkl) produces a diffraction cone of half-angle 2θ , forming rings on a detector.

The powder method is the most widely used technique for routine crystal identification. Each material produces a unique "fingerprint" of d -spacings and intensities, catalogued in the Powder Diffraction File (PDF) database containing over 1,000,000 entries.

Example: Aluminum powder XRD ($\text{Cu K}\alpha$, $\lambda = 1.5406 \text{ \AA}$):

(hkl)	2θ (deg)	d ($\text{\AA}$)	Relative Intensity
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(111)	38.47	2.338	very strong
(200)	44.74	2.024	strong
(220)	65.13	1.431	medium
(311)	78.23	1.221	medium
(222)	82.43	1.169	weak

The fcc selection rule (reflections only when h,k,l are all odd or all even) is directly confirmed.

2.2 Reciprocal Lattice

The reciprocal lattice vectors are:

[equation]

The Bragg condition is equivalent to the Laue condition: diffraction occurs when the scattering vector $\Delta \mathbf{k} = \mathbf{k}' - \mathbf{k}$ equals a reciprocal lattice vector $\mathbf{G}$.

Key results:

- * The reciprocal of sc (side a) is sc (side $2\pi/a$)
- * The reciprocal of bcc is fcc, and vice versa
- * The first Brillouin zone is the Wigner-Seitz cell of the reciprocal lattice

2.3 Structure Factor

The intensity of a diffracted beam is proportional to $|\mathbf{S}_{\mathbf{G}}|^2$, where:

[equation]

Structure Factor of the bcc Lattice

With atoms at (0,0,0) and (1/2, 1/2, 1/2):

[equation]

Structure Factor of the fcc Lattice

[equation]

These selection rules are the theoretical explanation for what Bragg observed experimentally in 1913: the systematic absences in KBr (fcc) vs. the complete pattern in KCl (appears sc).

Atomic Form Factor

The atomic form factor $f(\sin\theta/\lambda)$ equals the number of electrons Z at $\theta = 0$ and decreases with scattering angle. Hubbell et al. (1975) tabulated form factors for all elements $Z = 1$ to 100 -- one of the most cited works in X-ray physics with 1678 citations [Hubbell 1975].

Atom	$s = 0$	$s = 0.25 \text{ \AA}^{-1}$	$s = 0.50 \text{ \AA}^{-1}$	$s = 1.00 \text{ \AA}^{-1}$
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C (Z=6)	6.0	4.1	2.3	0.9
Na (Z=11)	11.0	7.8	4.7	2.0
Cl (Z=17)	17.0	12.4	7.8	3.5
Cu (Z=29)	29.0	22.5	15.2	7.4
Ge (Z=32)	32.0	25.1	17.2	8.5

2.4 Neutron Diffraction

Neutrons with wavelengths ~ 1 Å at thermal energies (E approx 25 meV) are also diffracted by crystals. However, neutrons scatter from nuclei rather than electron clouds, giving complementary information.

Shull and Wollan: Pioneers of Neutron Diffraction

Clifford Shull and Ernest Wollan at Oak Ridge National Laboratory developed neutron diffraction in the late 1940s, using neutron beams from the X-10 graphite reactor built during the Manhattan Project.

Their landmark results:

- * 1948: First neutron powder diffraction patterns, determining coherent neutron scattering lengths for many elements [Wollan & Shull 1948].
- * 1949: Structure of ice by neutron diffraction -- locating hydrogen atoms invisible to X-rays.
- * 1951: First experimental evidence of antiferromagnetic order in MnO -- magnetic Bragg peaks below the Néel temperature absent in X-ray diffraction [Shull, Strauser & Wollan 1951]. This paper has 1321 citations.

Shull described the early days in his 1995 Nobel lecture: "I would like to express... my deep regret that Ernest Wollan, who first guided me to the wonders of neutron-scattering, could not have shared these experiences." [Shull 1995, p. 753]

Shull received the 1994 Nobel Prize in Physics "for the development of the neutron diffraction technique." Wollan had died in 1984.

Advantages of neutrons over X-rays: 1. Locate light atoms (H, Li, O) near heavy atoms 2. Distinguish neighboring elements (Mn vs Fe) 3. Detect magnetic order through neutron magnetic moment interaction

2.5 Electron Diffraction: The Davisson-Germer Experiment (1927)

In one of the most important experiments of the 20th century, Clinton Davisson and Lester Germer at Bell Labs demonstrated the wave nature of electrons by observing their diffraction from a nickel crystal [Davisson & Germer 1927].

The Experiment

The experiment was partly accidental. While studying electron scattering from polycrystalline nickel in 1925, a liquid air bottle exploded in their laboratory, breaking the vacuum system. The nickel target oxidized. When heated to remove the oxide, the polycrystalline nickel recrystallized into a few large single crystals. When they resumed scattering measurements, they observed sharp diffraction peaks.

From the original 1927 Nature paper:

"In a series of experiments now in progress, we are directing a narrow beam of electrons normally against a target cut from a

single crystal of nickel, and are measuring the intensity of scattering... in various directions in front of the target." -- Davisson & Germer, Nature 119, 558 (1927)

The Key Data

Davisson and Germer's Figure 2 (p. 559) shows the intensity of electron scattering versus azimuth angle at 54 volts and co-latitude 50° . Three sharp spurs appear at 120° intervals, confirming the threefold symmetry of the Ni(111) surface.

"The spurs are due to beams of scattered electrons which are nearly if not quite as well defined as the primary beam." -- Davisson & Germer 1927, p. 558

Their Table I provides the quantitative verification of de Broglie's wave hypothesis. For the {111} azimuth:

Voltage (V)	Co-latitude theta	Electron speed v (10 ⁸ cm/s)	nlambda (Å)	n(lambda mv/h)
54	50°	4.36	1.65	0.99
100	31°	5.94	1.11	0.91
174	21°	7.84	0.77	0.83

For the {100} azimuth:

Voltage (V)	Co-latitude theta	Electron speed v (10 ⁸ cm/s)	nlambda (Å)	n(lambda mv/h)
65	44°	4.79	1.49	0.98
126	29°	6.67	1.04	0.95
190	20°	8.19	0.74	0.83

The last column n(lambda mv/h) should equal an integer if the de Broglie relation $\lambda = h/mv$ holds. The values cluster near 1.0 -- direct experimental proof that electrons behave as waves with wavelength $\lambda = h/mv$.

The plane spacings used in the analysis were:

- * {111} and {100}: $d = 2.15 \times 10^{-8}$ cm
- * {110}: $d = 1.24 \times 10^{-8}$ cm

Davisson and Germer noted: "These results are highly suggestive, of course, of the ideas underlying the theory of wave mechanics."

This experiment confirmed de Broglie's 1924 hypothesis. De Broglie received the 1929 Nobel Prize; Davisson shared the 1937 Nobel Prize with G. P. Thomson.

References

1. Bragg, W. L. (1913). "The Structure of Some Crystals as Indicated by Their Diffraction of X-rays." Proc. R. Soc. Lond. A 89, 248-277. DOI: 10.1098/rspa.1913.0083.
2. Davisson, C. and Germer, L. H. (1927). "The Scattering of Electrons by a Single Crystal of Nickel." Nature 119, 558-560. DOI: 10.1038/119558a0. -- The paper that proved the wave nature of electrons. Contains Fig. 1 (scattering setup), Fig. 2 (diffraction peaks at 54V), and Table I with the de Broglie verification data.
3. Shull, C. G., Strauser, W. A., and Wollan, E. O. (1951). "Neutron Diffraction by Paramagnetic and Antiferromagnetic Substances." Phys. Rev. 83, 333-345. DOI: 10.1103/PhysRev.83.333. -- First neutron diffraction evidence of antiferromagnetic order in MnO. 1321 citations.
4. Shull, C. G. (1995). "Early Development of Neutron Scattering." Rev. Mod. Phys. 67, 753-757. DOI: 10.1103/RevModPhys.67.753. -- Nobel Prize lecture.

5. Wollan, E. O. and Shull, C. G. (1948). "The Diffraction of Neutrons by Crystalline Powders." *Phys. Rev.* 73, 830. DOI: 10.1103/PhysRev.73.830. -- First neutron powder diffraction.
 6. Hubbell, J. H. et al. (1975). "Atomic Form Factors, Incoherent Scattering Functions, and Photon Scattering Cross Sections." *J. Phys. Chem. Ref. Data* 4, 471-538. DOI: 10.1063/1.555523. -- Form factors for $Z = 1-100$. 1678 citations.
 7. Pope, C. G. (1997). "X-ray Diffraction and the Bragg Equation." *J. Chem. Educ.* 74, 129. DOI: 10.1021/ed074p129. -- Pedagogical derivation of the Bragg equation.
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Chapter 3: Crystal Binding and Elastic Constants

Why do atoms form crystals? The answer lies in the lowering of total energy when atoms come together in an ordered arrangement. The nature of the binding force determines the crystal structure, the cohesive energy, and the elastic properties.

3.1 Crystals of Inert Gases

The noble gas solids (Ne, Ar, Kr, Xe) are the simplest crystals: bound entirely by the weak van der Waals interaction, all crystallizing in the fcc structure.

Van der Waals-London Interaction and the Lennard-Jones Potential

The total interaction is described by the Lennard-Jones potential:

[equation]

Boato and Casanova (1961) determined self-consistent LJ parameters for all noble gases from second virial coefficients, viscosity, and crystal properties. Anderson and Swenson (1975) measured the equations of state of solid Ar, Kr, and Xe under pressure, providing definitive cohesive energy and bulk modulus data [Anderson & Swenson 1975].

	ϵ/k_B (K)	σ (Å)	a (Å)	R_{nn} (Å)	E_{coh} (meV)	B (GPa)
Ne	35.7	2.789	4.464	3.156	20	1.1
Ar	119.8	3.405	5.311	3.755	80	2.7
Kr	164.0	3.624	5.646	3.992	116	3.5
Xe	230.0	3.963	6.129	4.334	170	3.6

$R_{\text{nn}} = a/\sqrt{2}$. LJ parameters from Boato & Casanova (1961); cohesive energies from Anderson & Swenson (1975).

3.2 Ionic Crystals

Madelung Energy

The dominant cohesive contribution in ionic crystals is the electrostatic (Madelung) energy:

[equation]

where α is the Madelung constant. Mestechkin (2000) tabulated over 200 Madelung constants [Mestechkin 2000].

Structure	Madelung constant α	Example
NaCl (B1)	1.7476	NaCl, MgO, LiF
CsCl (B2)	1.7627	CsCl, CsBr
Zinc blende (B3)	1.6381	ZnS, GaAs
Wurtzite (B4)	1.6413	ZnO, AlN
Fluorite	2.5194	CaF ₂ , UO ₂

Rutile	2.408	TiO?
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Born-Mayer Lattice Energies

The Born-Haber cycle (purely experimental thermodynamic data) agrees with the Born-Mayer electrostatic model to within ~1%:

Crystal	Expt. (kJ/mol)	Calc. (kJ/mol)	Diff. (%)
LiF	1037	1030	0.7
NaCl	786	778	1.0
KCl	715	709	0.8
KBr	682	674	1.2
RbCl	689	681	1.2
CsCl	657	649	1.2

3.3 Cohesive Energies Compared

Crystal	E_{coh} (eV/atom)	Bonding type
C (diamond)	7.37	covalent
Si	4.63	covalent
Ge	3.85	covalent
SiC	6.34	covalent
Na	1.11	metallic
Cu	3.49	metallic
Fe	4.28	metallic
W	8.90	metallic
NaCl	3.28	ionic (per ion pair)

3.4 Elastic Constants

Ultrasonic Measurement Method

The standard technique measures the velocity of ultrasonic waves along crystallographic directions. The key relations for cubic crystals:

- * [001] longitudinal: $\rho v^2 = c_{11}$
- * [001] transverse: $\rho v^2 = c_{44}$
- * [011] longitudinal: $\rho v^2 = (c_{11} + c_{12} + 2c_{44})/2$
- * [011] transverse (polarized [01 $\bar{1}$]): $\rho v^2 = (c_{11} - c_{12})/2$

The Overton-Gaffney Experiment (1955)

Overton and Gaffney performed the definitive measurement of copper's elastic constants from 4.2 K to 300 K using the ultrasonic

pulse-echo technique at 10 MHz [Overton & Gaffney 1955]. Their paper (636 citations) remains the standard reference for Cu elastic data.

From the original paper: "The ultrasonic pulse technique has been used in conjunction with a specially devised cryogenic technique to measure the velocities of 10-Mc/sec acoustic waves in copper single crystals in the range from 4.2°K to 300°K." [Overton & Gaffney 1955, p. 969]

Experimental details:

- * (001) copper crystal: length 5.3622 cm at 27.4°C
- * (011) copper crystal: length 5.7833 cm at 20°C
- * Ultrasonic frequency: 10 MHz
- * Precision: c_{44} to $\pm 0.25\%$, velocities to $\pm 0.08\%$

Elastic Constants of Cubic Crystals

Crystal	c_{11}	c_{12}	c_{44}	$A = 2c_{44}/(c_{11}-c_{12})$
Cu	16.84	12.14	7.54	3.21
Al	10.82	6.13	2.85	1.21
Au	18.60	15.70	4.20	2.90
Fe	23.10	13.50	11.60	2.42
W	52.33	20.45	16.07	1.01
NaCl	4.87	1.24	1.26	0.69
KCl	4.07	0.71	0.63	0.37
LiF	11.12	4.20	6.28	1.82
MgO	29.71	9.54	15.57	1.54
Si	16.58	6.39	7.96	1.56
Ge	12.89	4.83	6.71	1.66
Diamond	107.6	12.50	57.74	1.21

Units: 10^{11} N/m². Cu data from Overton & Gaffney (1955); alkali halides at 4.2 K from Lewis et al. (1967).

Anisotropy ratio A: tungsten is nearly isotropic ($A = 1.01$), copper is strongly anisotropic ($A = 3.21$). Overton and Gaffney noted: "The isotropy, $(c_{11}-c_{12})/2c_{44}$, was observed to remain practically constant from 4.2°K to 180°K." [Overton & Gaffney 1955, p. 969]

Bulk Modulus

[equation]

Material	B (GPa)	Notes
Diamond	443	hardest known
W	310	highest among metals
Fe	167	
Cu	137	
Si	98	
Al	76	
NaCl	24	
KCl	18	

Na	6.3	softest cubic metal
Xe (solid)	3.6	van der Waals

Slater (1924) performed the first compressibility measurements on alkali halides [Slater 1924], establishing the experimental basis that softer crystals have larger ions.

References

1. Anderson, M. S. and Swenson, C. A. (1975). "Experimental equations of state for the rare gas solids." J. Phys. Chem. Solids 36, 145. DOI: 10.1016/0022-3697(75)90004-990004-9).
2. Overton, W. C. Jr. and Gaffney, J. (1955). "Temperature Variation of the Elastic Constants of Cubic Elements. I. Copper." Phys. Rev. 98, 969-977. DOI: 10.1103/PhysRev.98.969. -- Definitive Cu elastic constants from 4.2 K to 300 K. 636 citations.
3. Lewis, J. T., Lehoczy, A., and Briscoe, C. V. (1967). "Elastic Constants of the Alkali Halides at 4.2 K." Phys. Rev. 161, 877. DOI: 10.1103/PhysRev.161.877. -- Complete elastic constants of all alkali halides. 331 citations.
4. Slater, J. C. (1924). "Compressibility of the Alkali Halides." Phys. Rev. 23, 488. DOI: 10.1103/PhysRev.23.488. -- First compressibility measurements.
5. Schwerdtfeger, P., Gaston, N., and Krawczyk, R. P. (2006). "Extension of the Lennard-Jones potential..." Phys. Rev. B 73, 064112. DOI: 10.1103/PhysRevB.73.064112.
6. Mestechkin, M. M. (2000). "Electrostatic parameters of ionic crystals." J. Phys. Chem. Ref. Data 29, 571.

Part II: Lattice Dynamics and Thermal Properties

Chapter 4: Phonons I -- Crystal Vibrations

The atoms in a crystal vibrate about their equilibrium positions. These vibrations, quantized as phonons, determine most thermal, acoustic, and transport properties. This chapter presents the experimental phonon dispersion curves that reveal how lattice vibrations propagate through crystals.

4.1 Vibrations of Crystals with Monatomic Basis

For a 1D chain of identical atoms (mass M , spacing a , force constant C):

[equation]

Key features:

- * Long wavelengths ($k \rightarrow 0$): $\omega \approx v_s |k|$, where $v_s = \sqrt{C/M}$
- * Zone boundary ($k = \pi/a$): $\omega_{\max} = 2\sqrt{C/M}$, group velocity $v_g = 0$
- * Periodicity: first Brillouin zone is $-\pi/a < k \leq \pi/a$

4.2 The First Phonon Dispersion Curve: Brockhouse and Stewart (1955)

The experimental measurement of phonon dispersion relations was made possible by inelastic neutron scattering, developed by Bertram Brockhouse at Chalk River, Canada. Brockhouse received the 1994 Nobel Prize in Physics.

The Experiment

From the original paper: "Monoenergetic neutrons provided by a crystal spectrometer (wavelength 1.14 Å, energy 0.062 eV) were scattered at an angle of 95.1° by an aluminum single crystal." [Brockhouse & Stewart 1955, p. 757]

The method works by conservation of energy and momentum:

[equation]

[equation]

By measuring the energy and direction of scattered neutrons, the phonon frequency $\omega(\mathbf{q})$ is determined.

Brockhouse's Data: Aluminum Phonon Dispersion

Fig. 3 of Brockhouse & Stewart (1955) shows the first phonon dispersion curve ever measured -- ω vs q for aluminum [Brockhouse & Stewart 1955, p. 757]:

- * Longitudinal branch (open circles): rising to $\sim 6 \times 10^{13}$ rad/sec
- * Transverse branch (filled circles, near $\{111\}$ axes): rising to $\sim 3 \times 10^{13}$ rad/sec
- * The dashed curve is a sinusoidal fit for a 1D model

The transverse sound velocity extracted from the low- q slope: $v_T = 3080$ m/sec, in agreement with the elastic constants of Chapter 3.

Precision Measurements: Stedman and Nilsson (1966)

Stedman and Nilsson performed the most precise Al phonon measurements at 80 K and 300 K (204 citations), achieving errors of only ~0.6% [Stedman & Nilsson 1966]. Key findings:

- * Kohn anomalies in the dispersion curves (abrupt changes caused by electronic screening at 2k_F)
- * Phonon widths at 80 K from the electron-phonon interaction
- * Force constants extending to 15th neighbors (range: 2.6 lattice spacings)

Phonon Frequencies at High-Symmetry Points in Aluminum

Symmetry point	Branch	Frequency (THz)
X [100] zone boundary	L	9.69
X [100] zone boundary	T	5.76
L [111] zone boundary	L	9.78
L [111] zone boundary	T	4.48
K [110] zone boundary	T ₁	7.44

From Brockhouse (1958).

4.3 Two Atoms per Primitive Basis: NaI

When the primitive cell contains two atoms, the dispersion splits into acoustic (atoms in phase) and optical (atoms out of phase) branches.

Woods, Cochran, and Brockhouse (1960): NaI Phonon Dispersion

The landmark demonstration of acoustic and optical branches was by Woods et al. (1960), who measured the complete phonon dispersion of NaI [Woods 1960]. NaI was chosen because the large mass ratio ($M_{\text{I}}/M_{\text{Na}} = 127/23 = 5.5$) produces a wide gap between branches.

With 2 atoms per primitive cell: 6 branches total (1 LA + 2 TA + 1 LO + 2 TO).

Direction	Branch	Gamma point (THz)	Zone boundary (THz)
[100]	LA	0	2.7
[100]	TA	0	1.8
[100]	LO	4.5	3.8
[100]	TO	3.6	2.1

From Woods et al. (1960), Phys. Rev. 119, 980.

The frequency gap between the acoustic branches (~2.7 THz) and optical branches (~3.6 THz) is clearly visible. Electromagnetic radiation in this range cannot couple to the lattice -- the Reststrahlen band.

4.4 Quantization and Phonon Momentum

A phonon has energy $\hbar\omega$ and crystal momentum $\hbar\mathbf{q}$. The mean occupation at temperature T :
[equation]

Crystal momentum is defined only modulo $\mathbf{G}$ -- this distinction matters for Umklapp processes (Chapter 5).

References

1. Brockhouse, B. N. and Stewart, A. T. (1955). "Scattering of Neutrons by Phonons in an Aluminum Single Crystal." *Phys. Rev.* 100, 756-757. DOI: 10.1103/PhysRev.100.756. -- The first phonon dispersion curve ever measured. Contains Fig. 3 showing ω vs q for Al with transverse sound velocity 3080 m/sec.
 2. Brockhouse, B. N. (1958). "Normal Modes of Aluminum by Neutron Spectrometry." *Rev. Mod. Phys.* 30, 236. DOI: 10.1103/RevModPhys.30.236.
 3. Woods, A. D. B., Cochran, W., and Brockhouse, B. N. (1960). "Lattice Dynamics of Alkali Halide Crystals." *Phys. Rev.* 119, 980-999. DOI: 10.1103/PhysRev.119.980. -- NaI phonon dispersion showing acoustic + optical branches. 654 citations.
 4. Woods, A. D. B. et al. (1963). "Lattice Dynamics of Alkali Halide Crystals. II." *Phys. Rev.* 131, 1025. DOI: 10.1103/PhysRev.131.1025.
 5. Stedman, R. and Nilsson, G. (1966). "Dispersion Relations for Phonons in Aluminum at 80 and 300 K." *Phys. Rev.* 145, 492. DOI: 10.1103/PhysRev.145.492. -- Precision Al phonon data. 204 citations.
 6. Kresch, M. et al. (2008). "Phonons in aluminum at high temperatures studied by inelastic neutron scattering." *Phys. Rev. B* 77, 024301. -- Al phonon DOS at 10, 150, 300, 525, 775 K.
-

Chapter 5: Phonons II -- Thermal Properties

The thermal properties of solids -- heat capacity, thermal conductivity, thermal expansion -- are governed by lattice vibrations. The experimental data in this chapter demonstrate how phonon statistics and scattering processes determine these properties across an enormous temperature range.

5.1 Phonon Heat Capacity

The Debye Model and the T^3 Law

At low temperatures:

[equation]

where Θ_D is the Debye temperature: $k_B \Theta_D = \hbar \omega_D$.

The standard test is to plot C/T vs T^2 at low temperatures -- the data should be linear:

[equation]

where γ is the electronic contribution (Ch. 6) and β gives Θ_D . Parkinson (1958) performed this analysis for many metals, showing that the Debye model fits remarkably well below $\Theta_D/10$ [Parkinson 1958].

Debye Temperatures

Material	Θ_D (K)	Material	Θ_D (K)
C (diamond)	2230	Cu	343
Si	645	Au	165
Ge	374	Al	428
SiC	1200	Fe	470
MgO	946	W	400
NaCl	321	Na	158
KCl	235	Pb	105
LiF	732	Ar (solid)	93

From Abrahams et al. (1975) and Grimvall et al. (1974).

Diamond has the highest Θ_D of any material (2230 K). At room temperature ($T/\Theta_D \approx 0.13$), diamond is far from the classical limit -- this is why Einstein's 1907 quantum theory of specific heat was inspired by diamond's anomalously low heat capacity.

Einstein Model

Einstein (1907) treated all atoms as independent oscillators at frequency ω_E :

[equation]

The Einstein model correctly predicts the exponential freeze-out at $T \ll \Theta_E$ but gives $C \propto e^{-\Theta_E/T}$ instead of the

correct T^3 . The Debye model corrects this by accounting for the distribution of phonon frequencies.

5.2 Thermal Conductivity

The Phonon Gas Model

In a nonmetallic crystal:

[equation]

The temperature dependence reflects three competing scattering mechanisms: 1. Boundary scattering (low T): $\ell \propto$ approx crystal size, $\kappa \propto T^3$ 2. Point defect scattering: $\ell \propto \omega^{-4}$ (Rayleigh) 3. Umklapp scattering (high T): $\ell \propto e^{-\Theta_D/bT}$, κ decreases

The result is a characteristic thermal conductivity peak at $T \sim \Theta_D/20$.

Diamond: The Highest Thermal Conductivity

Diamond has the highest κ of any bulk material. Inyushkin et al. (2018) measured κ of high-purity synthetic single-crystal diamonds from 6 K to 400 K, separating normal and Umklapp phonon-phonon scattering contributions [Inyushkin 2018]:

T (K)	κ (W/m·K)
6	15
10	120
20	2800
40	12000
80	14000
150	4500
300	2200
400	1600

From Inyushkin et al. (2018), Phys. Rev. B 97, 144305.

The enormous peak at 80 K ($\kappa = 14,000$ W/m·K) and subsequent rapid decrease demonstrate Umklapp scattering onset. At room temperature, κ approx 2200 W/m·K -- still 5x higher than copper.

Umklapp Processes

In a normal (N) process: $\mathbf{q}_1 + \mathbf{q}_2 = \mathbf{q}_3$ (momentum conserved within the zone).

In an Umklapp (U) process: $\mathbf{q}_1 + \mathbf{q}_2 = \mathbf{q}_3 + \mathbf{G}$ (a reciprocal lattice vector is needed).

Only U-processes create thermal resistance. Their rate decreases exponentially at low T ($\propto e^{-\Theta_D/bT}$) because they require zone-boundary phonons that are thermally activated.

Room-Temperature Thermal Conductivity

Material	κ (W/m·K)	Dominant carrier
Diamond	2200	phonons

Cu	401	electrons
Al	237	electrons
Si	148	phonons
Fe	80	electrons + phonons
GaAs	46	phonons
NaCl	7	phonons
Glass (SiO ₂)	1.4	phonons (localized)

References

1. Inyushkin, A. V. et al. (2018). "Thermal conductivity of high purity synthetic single crystal diamonds." Phys. Rev. B 97, 144305. DOI: 10.1103/PhysRevB.97.144305. -- Diamond $\kappa(T)$ from 6 K to 400 K. Peak 14000 W/m·K at 80 K.
2. Parkinson, D. H. (1958). "The specific heats of metals at low temperatures." Rep. Prog. Phys. 21, 307. DOI: 10.1088/0034-4885/21/1/307. -- Comprehensive review of metal specific heat with Debye fits.
3. Abrahams, S. C. et al. (1975). "Debye temperatures and cohesive properties." J. Chem. Phys. 63, 1162.
4. Grimvall, G. et al. (1974). "Correlation of properties of materials to Debye and melting temperatures." Phys. Scripta 10, 340.

Part III: Electrons in Solids

Chapter 6: Free Electron Fermi Gas

The free electron model treats valence electrons in a metal as non-interacting fermions confined to the crystal volume. Despite its simplicity, it explains the electronic heat capacity, electrical and thermal conductivity, and the Hall effect with remarkable accuracy.

6.1 Energy Levels and the Fermi-Dirac Distribution

At $T = 0$, all states up to the Fermi energy E_F are occupied:

[equation]

Metal	n (10^{22} m^{-3})	E_F (eV)	T_F (10^4 K)	v_F (10^7 m/s)	k_F ($\text{\AA}^{-1}$)
Li	4.70	4.74	5.51	1.29	1.12
Na	2.65	3.24	3.77	1.07	0.92
K	1.40	2.12	2.46	0.86	0.75
Cu	8.47	7.04	8.16	1.57	1.36
Ag	5.86	5.49	6.38	1.39	1.20
Au	5.90	5.53	6.42	1.40	1.21

Fermi temperatures are $\sim 10^4 \text{ K}$ -- far above room temperature. Only electrons near E_F participate in thermal and transport processes.

6.2 Heat Capacity of the Electron Gas

The electronic heat capacity: $C_{el} = \gamma T$, where γ is the Sommerfeld coefficient.

Martin's Precision Measurements (1968)

Douglas L. Martin at the National Research Council of Canada measured the specific heat of pure Cu, Ag, and Au below 3 K with extraordinary precision [Martin 1968].

From the original paper: "The electronic specific-heat coefficient γ and the limiting low-temperature Debye temperature (Θ_D) are estimated as $165.2 \pm 0.8 \text{ mJ/mol} \cdot \text{K}^2$ (g atom) and $345.8 \pm 1.2 \text{ K}$ for copper, $153.1 \pm 0.9 \text{ mJ/mol} \cdot \text{K}^2$ (g atom) and $227.3 \pm 0.6 \text{ K}$ for silver, and $165.1 \pm 1.0 \text{ mJ/mol} \cdot \text{K}^2$ (g atom) and $162.3 \pm 0.5 \text{ K}$ for gold." [Martin 1968, p. 650]

The data were fitted to: $C = \gamma T + [464.34/(\Theta_D^3)]T^3$ (Table II, p. 652).

Sample purity (Table I, p. 651): Cu sample 140.663 g with Fe < 0.2 ppm, Ag 134.271 g with Fe = 0.43 ppm, Au 198.887 g with Fe = 0.16 ppm -- all vacuum-cast to avoid oxide contamination.

Metal	γ_{exp} ($\text{mJ/mol} \cdot \text{K}^2$)	γ_{free} ($\text{mJ/mol} \cdot \text{K}^2$)	m^*/m_e	Θ_D (K)
Li	1.63	0.749	2.18	--
Na	1.38	1.094	1.26	--
K	2.08	1.668	1.25	--
Cu	0.695	0.505	1.38	345.8

Ag	0.646	0.645	1.01	227.3
Au	0.729	0.642	1.14	162.3
Al	1.35	0.912	1.48	--

Bold values: directly from Martin (1968), Table II. $m^*/m_e = \gamma_{\text{exp}}/\gamma_{\text{free}}$.*

Silver is nearly free-electron-like ($m^*/m_e = 1.01$). Martin noted that "the gamma value for gold is significantly lower than previously reported values"* -- his improved calorimetry resolved discrepancies from earlier work.

Heavy Fermions

In certain rare-earth compounds, gamma is enormous:

Compound	gamma (mJ/mol·K ²)	Notes
CeAl?	1620	prototypical heavy fermion
CeCu ₂ Si?	1000	first heavy fermion superconductor
UBe??	1100	
Cu (comparison)	0.695	normal metal

The effective mass in CeAl? is $m^* \approx 1000 m_e$.

6.3 Electrical Resistivity

The Matula Reference Data (1979)

Matula compiled the definitive recommended resistivity values for Cu, Au, Pd, and Ag from cryogenic to beyond the melting point -- a 904-citation NIST reference paper [Matula 1979].

T (K)	rho (muOhm·cm)
4	0.002
10	0.002
20	0.008
40	0.054
60	0.162
80	0.322
100	0.520
150	0.963
200	1.358
250	1.524
300	1.678
400	2.402
500	3.090
600	3.792
800	5.262
1000	6.858

From Matula (1979), J. Phys. Chem. Ref. Data 8, 1147.

Residual resistivity ratio $RRR = \rho(300K)/\rho(4K)$ approx 840. Below ~40 K: Bloch-Grüneisen T^5 behavior; above ~150 K: nearly linear in T .

6.4 Hall Effect

The Hall coefficient: $R_H = E_y/(j_x B_z) = -1/(ne)$ for free electrons.

Metal	R_H (exp) ($10^{-11} \text{ m}^3/\text{C}$)	$-1/ne$ (free)	Sign
Li	-1.7	-1.3	- (electron)
Na	-2.5	-2.4	-
K	-4.2	-4.5	-
Cu	-0.55	-0.74	-
Ag	-0.84	-1.1	-
Au	-0.72	-1.1	-
Al	-0.30	-0.35	-
Be	+2.4	--	+ (hole!)
Zn	+0.33	--	+
Cd	+0.60	--	+

From Hurd (2012), The Hall Effect in Metals and Alloys (1350 citations).

Be, Zn, and Cd have positive Hall coefficients -- indicating hole-like carriers. This anomaly requires band theory (Ch. 7).

6.5 Wiedemann-Franz Law

[equation]

Metal	L ($10^{-8} \text{ W}\cdot\text{Ohm}/\text{K}^2$)
Cu	2.23
Ag	2.31
Au	2.35
Al	2.14
Fe	2.61
W	3.04
Pb	2.47

At 273 K. Theoretical: $L_0 = 2.44$.

Agreement within ~10%. Völklein et al. (2009) showed the Wiedemann-Franz law holds even for individual metallic nanowires [Völklein 2009].

References

1. Martin, D. L. (1968). "Specific Heats below 3°K of Pure Copper, Silver, and Gold, and of Extremely Dilute Gold-Transition-Metal Alloys." *Phys. Rev.* 170, 650-655. DOI: 10.1103/PhysRev.170.650. -- γ and Θ_D for Cu, Ag, Au with sub-ppm purity. Table I: sample weights and impurities. Table II: fitted coefficients with 95% confidence limits.
 2. Matula, R. A. (1979). "Electrical Resistivity of Copper, Gold, Palladium, and Silver." *J. Phys. Chem. Ref. Data* 8, 1147-1298. DOI: 10.1063/1.555614. -- Recommended Cu resistivity from cryogenic to liquid. 904 citations.
 3. Hurd, C. (2012). *The Hall Effect in Metals and Alloys*. Springer. -- Comprehensive Hall coefficient compilation. 1350 citations.
 4. Völklein, F. et al. (2009). "The experimental investigation of thermal conductivity and the Wiedemann-Franz law for single metallic nanowires." *Nanotechnology* 20, 325706. DOI: 10.1088/0957-4484/20/32/325706.
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Chapter 7: Energy Bands

The free electron model fails to explain why some materials are metals and others insulators. The resolution: solving the Schrödinger equation for electrons in the periodic potential of the crystal lattice produces energy bands separated by forbidden gaps.

7.1 Nearly Free Electron Model

A weak periodic potential $V(\mathbf{r})$ opens a band gap at each Brillouin zone boundary:

[equation]

States at the zone boundary form standing waves -- one with density peaked on ions (lower energy), the other between ions (higher energy).

7.2 Bloch's Theorem

Felix Bloch (1928) proved that electron wavefunctions in a periodic potential have the form:

[equation]

where $u_{\mathbf{k}}$ has the periodicity of the lattice.

7.3 Metals vs Insulators

- * Metal: partially filled band -- electrons at E_F can be accelerated
- * Insulator: filled bands separated by large gap ($E_g > 4$ eV)
- * Semiconductor: small gap ($E_g \sim 0.1-3$ eV), allowing thermal excitation

7.4 Experimental Band Structure: ARPES

ARPES (angle-resolved photoemission spectroscopy) directly measures $E(\mathbf{k})$:

[equation]

Asonen et al. (1982) mapped the band structure of Cu alloy surfaces along (100), (110), (111), resolving the d-bands and sp-band [Asonen 1982].

7.5 Band Gap Temperature Dependence

Pässler (1999) compiled Varshni parameters for the empirical formula [Pässler 1999]:

[equation]

Material	$E_g(0)$ (eV)	Type	α (10^{-4} eV/K)	β (K)	$E_g(300K)$ (eV)
Si	1.170	indirect	4.73	636	1.12
Ge	0.744	indirect	4.77	235	0.66
GaAs	1.519	direct	5.41	204	1.42

InP	1.424	direct	4.50	327	--
GaP	2.338	indirect	6.20	460	--
InSb	0.235	direct	3.20	170	0.17

From Pässler (1999), Phys. Status Solidi B 216, 975.

7.6 Number of Orbitals in a Band

Each band holds $2N$ electrons (N primitive cells, 2 spins). Elements with an odd number of electrons per cell (Na, Cu, Al) are always metallic. Even-electron elements can be insulators if the gap is large enough (C, Si, Ge).

References

1. Pässler, R. (1999). "Parameter Sets Due to Fittings of the Temperature Dependencies of Fundamental Bandgaps in Semiconductors." Phys. Status Solidi B 216, 975. DOI: 10.1002/(SICI)1521-3951(199912)216:2<975::AID-PSSB975>3.0.CO;2-N1521-3951(199912)216:2<975::AID-PSSB975>3.0.CO;2-N).
2. Asonen, H. et al. (1982). "ARPES study of (100), (110), (111) surfaces of Cu₂Al₂Si₂." Phys. Rev. B 25, 7075. DOI: 10.1103/PhysRevB.25.7075.

Chapter 8: Semiconductor Crystals

Semiconductors are the foundation of modern electronics. Their tunable band gaps, controllable carrier concentrations, and high mobilities have enabled the transistor, IC, laser, and solar cell.

8.1 Effective Mass: Cyclotron Resonance

The Dresselhaus-Kip-Kittel Experiment (1955)

The most direct measurement of effective mass is cyclotron resonance: carriers in a magnetic field orbit at frequency $\omega_c = eB/m^*$. By sweeping the microwave frequency or field, m^* is determined directly.

Dresselhaus, Kip, and Kittel at UC Berkeley performed the definitive cyclotron resonance experiments on Si and Ge at 4 K [Dresselhaus 1955]:

"An experimental and theoretical discussion is given of the results of cyclotron resonance experiments on charge carriers in silicon and germanium single crystals near 4°K." -- Dresselhaus, Kip, Kittel, Phys. Rev. 98, 368 (1955)

Experimental conditions: Microwave frequencies 9000 and 24,000 MHz; temperature 4 K (liquid helium); magnetic field swept 0-10,000 Oersteds. The condition for observing resonance: $\omega_c \tau \geq 1$, requiring $\tau \approx 10^{-11}$ sec.

Fig. 2 of the original paper (p. 372) shows the actual cyclotron resonance absorption spectrum of Ge -- distinct peaks for electrons and holes at different magnetic field values.

Effective Masses (from the abstract, p. 368)

Germanium -- electron energy surfaces along $\langle 111 \rangle$ are prolate spheroids:

* Longitudinal mass: $m_l = (1.58 \pm 0.04) m_e$

* Transverse mass: $m_t = (0.082 \pm 0.001) m_e$

Silicon -- electron energy surfaces along $\langle 100 \rangle$ are prolate spheroids:

* Longitudinal mass: $m_l = (0.97 \pm 0.02) m_e$

* Transverse mass: $m_t = (0.19 \pm 0.01) m_e$

The energy surface equation:

[equation]

with constants (from the abstract):

* Ge: $A = -(13.0 \pm 0.2) \hbar^2/2m$, $|B| = (8.9 \pm 0.1) \hbar^2/2m$, $|C| = (10.3 \pm 0.2) \hbar^2/2m$

* Si: $A = -(4.1 \pm 0.2) \hbar^2/2m$, $|B| = (1.6 \pm 0.2) \hbar^2/2m$, $|C| = (3.3 \pm 0.5) \hbar^2/2m$

Complete Effective Mass Table

Material	m_l^*/m_e	m_t^*/m_e	m_{hh}^*/m_e	m_{lh}^*/m_e	E_g (eV)
Si	0.97 $\pm$ 0.02	0.19 $\pm$ 0.01	0.49	0.16	1.12
Ge	1.58 $\pm$ 0.04	0.082 $\pm$ 0.001	0.33	0.04	0.66
GaAs	0.067	--	0.45	0.082	1.42

InSb	0.014	--	0.40	0.016	0.24
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Bold: directly from Dresselhaus, Kip, Kittel (1955).

8.2 Intrinsic Carrier Concentration

[equation]

Material	n_i at 300 K (cm ⁻³)	E_g (eV)
Si	1.0×10^{10}	1.12
Ge	2.4×10^{13}	0.66
GaAs	1.8×10^6	1.42
InSb	1.6×10^{17}	0.24

The exponential dependence on $E_g/k_B T$ spans 10 orders of magnitude.

8.3 Donor and Acceptor Ionization Energies

Dean et al. (1967) measured ionization energies from radiative recombination involving neutral donors and acceptors in Si and Ge [Dean 1967]:

Type	Impurity	In Si (meV)	In Ge (meV)
Donor	P	45	12.0
Donor	As	54	12.7
Donor	Sb	39	9.6
Acceptor	B	45	10.4
Acceptor	Al	67	10.2
Acceptor	Ga	72	10.8
Acceptor	In	160	11.2

From Dean et al. (1967), Phys. Rev. 161, 711.

8.4 Carrier Mobility

Li and Thurber (1977) measured electron mobility in P-doped Si [Li & Thurber 1977]:

N_D (cm ⁻³)	μ_e (cm ² /V·s)
10^{14}	1450
10^{15}	1350
10^{16}	1200
10^{17}	750
10^{18}	280
10^{19}	110

At low doping: phonon scattering dominates ($\mu \approx 1450$). At high doping: ionized impurity scattering (μ drops 13x).

References

1. Dresselhaus, G., Kip, A. F., and Kittel, C. (1955). "Cyclotron Resonance of Electrons and Holes in Silicon and Germanium Crystals." *Phys. Rev.* 98, 368-384. DOI: 10.1103/PhysRev.98.368. -- Effective masses of Si and Ge from cyclotron resonance at 4 K. Fig. 2: actual absorption spectrum.
 2. Dean, P. J. et al. (1967). "New radiative recombination processes involving neutral donors and acceptors in silicon and germanium." *Phys. Rev.* 161, 711. DOI: 10.1103/PhysRev.161.711.
 3. Li, S. S. and Thurber, W. R. (1977). "The dopant density and temperature dependence of electron mobility and resistivity in n-type silicon." *Solid-State Electron.* 20, 609. DOI: 10.1016/0038-1101(77)90100-990100-9).
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Chapter 9: Fermi Surfaces and Metals

The Fermi surface separates occupied from unoccupied electron states at $T = 0$. Its shape determines the electronic, thermal, and magnetic properties of metals.

9.1 The de Haas-van Alphen Effect

The most powerful probe of Fermi surface geometry: oscillations of magnetic susceptibility as a function of $1/B$. The Onsager relation connects oscillation period to the extremal cross-sectional area A :

[equation]

9.2 The Fermi Surface of Copper

Pippard (1957): First Fermi Surface Determination

A. B. Pippard used the anomalous skin effect to determine the Cu Fermi surface shape -- the first for any metal [Pippard 1957]. By measuring surface impedance vs crystallographic orientation, he showed Cu has a sphere with "necks" toward the L points.

Shoenberg (1962): Comprehensive dHvA Measurements

D. Shoenberg performed comprehensive dHvA measurements on Cu, Ag, and Au single crystals, identifying two principal oscillation frequencies [Shoenberg 1962]:

Orbit	Frequency (10 ⁷ G)	A/A_{free}
Belly <111>	5.90	1.00
Belly <100>	5.98	1.01
Belly <110>	5.88	1.00
Neck <111>	0.22	0.037

From Shoenberg (1962), Phil. Trans. R. Soc. A 255, 85.

The belly is nearly isotropic and close to free-electron value (ratio ~ 1.00). The neck is small ($\sim 4\%$ of belly), corresponding to narrow tubes at the 8 L-points. This sphere-with-8-necks topology explains Cu's excellent conductivity, its color (interband transitions at ~ 2 eV), and anomalous Hall effects in some orientations.

References

1. Shoenberg, D. (1962). "The Fermi surfaces of copper, silver and gold. I. The de Haas-Van Alphen effect." Phil. Trans. R. Soc. A 255, 85-133. DOI: 10.1098/rsta.1962.0011.
2. Pippard, A. B. (1957). "An experimental determination of the Fermi surface in copper." Phil. Trans. R. Soc. A 250, 325-357. DOI: 10.1098/rsta.1957.0023.

Part IV: Superconductivity

Chapter 10: Superconductivity

In 1911, Kamerlingh Onnes discovered that mercury's resistance drops to zero below 4.2 K. This phenomenon -- superconductivity -- is a macroscopic quantum state with no classical analog.

10.1 Discovery (Onnes, 1911)

Van Delft and Kes (2010) describe the historic moment: Onnes's notebook records "Mercury practically zero" [Van Delft 2010]. The transition is sharp: resistance drops by at least 10^{10} within ~ 0.01 K.

Transition Temperatures of Elements

Element	T _c (K)	Element	T _c (K)	Element	T _c (K)
Al	1.18	In	3.41	Pb	7.20
Cd	0.52	La	6.00	Sn	3.72
Ga	1.08	Hg	4.15	V	5.40
Hf	0.13	Mo	0.92	Zn	0.85
Nb	9.25	Os	0.66	Zr	0.61

Nb has the highest elemental T_c (9.25 K). Noble metals (Cu, Ag, Au) and ferromagnets (Fe, Co, Ni) are not superconducting.

10.2 Energy Gap: Giaever Tunneling (1960)

Ivar Giaever measured I-V characteristics of Al/AlO_x/Pb tunnel junctions, directly revealing the energy gap [Giaever 1974 Nobel lecture]:

Element	T _c (K)	2Δ(0) (meV)	2Δ(0)/k _B T _c
Al	1.18	0.36	3.53
Sn	3.72	1.15	3.59
In	3.41	1.05	3.57
Pb	7.20	2.73	4.40
Hg	4.15	1.66	4.64
Nb	9.25	3.05	3.82

BCS weak-coupling: 2Δ(0)/k_BT_c = 3.528. Al, Sn, In are weak-coupling; Pb and Hg are strong-coupling.

10.3 Josephson Effect: Shapiro Steps (1963)

Shapiro observed quantized voltage steps when microwaves irradiate a Josephson junction [Shapiro 1963]:

[equation]

This provides the most precise voltage standard in metrology. (1595 citations)

10.4 High-Temperature Superconductors

Bednorz and Müller discovered $T_c = 35$ K in $\text{La}_{1-x}\text{Ba}_x\text{CuO}_2$ in 1986 [Bednorz & Müller 1988 Nobel]:

Compound	T_c (K)	Year
$\text{La}_{1-x}\text{Ba}_x\text{CuO}_2$	35	1986
$\text{YBa}_2\text{Cu}_3\text{O}_{7-x}$	93	1987
$\text{Bi}_2\text{Sr}_2\text{Ca}_2\text{Cu}_3\text{O}_{10-x}$	110	1988
$\text{HgBa}_2\text{Ca}_2\text{Cu}_3\text{O}_{8-x}$	133	1993
H_2S (155 GPa)	203	2015
LaH_{10} (170 GPa)	250	2019

10.5 Type II Superconductors

Abrikosov predicted the vortex lattice (1957); Essmann and Träuble imaged it by Bitter decoration (1967) [Abrikosov 2004 Nobel].

10.6 Flux Quantization

Deaver and Fairbank (1961) showed flux through a superconducting ring is quantized: $\Phi_0 = h/2e = 2.068 \times 10^{-15}$ Wb. The factor of 2 confirms Cooper pairing.

References

1. Van Delft, D. and Kes, P. (2010). "The discovery of superconductivity." *Physics Today* 63(9), 38. DOI: 10.1063/1.3490499.
2. Giaever, I. (1974). "Electron Tunneling and Superconductivity." *Science* 183, 1253. DOI: 10.1126/science.183.4131.1253. -- Nobel lecture.
3. Shapiro, S. (1963). "Josephson Currents in Superconducting Tunneling." *Phys. Rev. Lett.* 11, 80. DOI: 10.1103/PhysRevLett.11.80. -- 1595 citations.
4. Bednorz, J. G. and Müller, K. A. (1988). "Perovskite-Type Oxides--The New Approach to High- T_c Superconductivity." *Rev. Mod. Phys.* 60, 585. DOI: 10.1103/RevModPhys.60.585. -- Nobel lecture.
5. Abrikosov, A. A. (2004). "Nobel Lecture: Type-II superconductors and the vortex lattice." *Rev. Mod. Phys.* 76, 975. DOI: 10.1103/RevModPhys.76.975.

Part V: Magnetism and Magnetic Resonance

Chapter 11: Diamagnetism and Paramagnetism

All materials respond to magnetic fields. Diamagnets are repelled (negative susceptibility); paramagnets are attracted (positive susceptibility).

11.1 Diamagnetic Susceptibility

Species	$\chi_{\text{dia}} (10^{-3} \text{ cm}^3/\text{mol})$	Species	$\chi_{\text{dia}} (10^{-3} \text{ cm}^3/\text{mol})$
He	-1.9	Cu?	-12
Ne	-7.2	Ge	-76
Ar	-19.3	Si	-13
Kr	-28.0	NaCl	-30
Xe	-43.0	Diamond	-5.9

11.2 Paramagnetism: Rare Earth Ions

The Curie law: $\chi = C/T = N\mu_0 p^2 \mu_B^2 / 3k_B T$, where $p = g\sqrt{J(J+1)}$.

Penney and Schlapp (1932) provided the first theoretical treatment of crystal field effects on rare earth susceptibilities [Penney & Schlapp 1932]. From the original paper: the crystal field potential $D(x^4+y^4+z^4)$ produces splittings of 389 cm⁻¹ for Pr and 834 cm⁻¹ for Nd in hydrated sulphates.

Guertin et al. (1973) measured effective moments of RE ions in Pd [Guertin 1973]:

Ion	Config.	J	g	p_{calc}	p_{exp}
Ce ³⁺	4f ¹	5/2	6/7	2.54	2.4
Pr ³⁺	4f ²	4	4/5	3.58	3.5
Nd ³⁺	4f ³	9/2	8/11	3.62	3.5
Sm ³⁺	4f ⁵	5/2	2/7	0.84	1.5
Gd ³⁺	4f ⁷	7/2	2	7.94	8.0
Dy ³⁺	4f ⁹	15/2	4/3	10.63	10.6
Er ³⁺	4f ¹¹	15/2	6/5	9.59	9.5
Yb ³⁺	4f ¹³	7/2	8/7	4.54	4.5

Agreement is remarkable except for Sm³⁺ (Van Vleck paramagnetism due to small J multiplet spacing).

11.3 Pauli Paramagnetism of Conduction Electrons

[equation]

Much smaller than Curie paramagnetism -- only electrons within $\sim k_B T$ of E_F can flip spin. Van Vleck (1953) reviewed the quantum theory comprehensively [Van Vleck 1953].

References

1. Penney, W. G. and Schlapp, R. (1932). "The Influence of Crystalline Fields on the Susceptibilities of Salts of Paramagnetic Ions. I. The Rare Earths." *Phys. Rev.* 41, 194. DOI: 10.1103/PhysRev.41.194. -- Crystal field splittings: Pr 389 cm⁻¹, Nd 834 cm⁻¹.
 2. Guertin, R. P. et al. (1973). "Magnetic moment, susceptibility, and electrical resistivity of dilute paramagnetic palladium-Rare-earth alloys." *Phys. Rev. B* 7, 274. DOI: 10.1103/PhysRevB.7.274.
 3. Van Vleck, J. H. (1953). "Models of Exchange Coupling in Ferromagnetic Media." *Rev. Mod. Phys.* 25, 220. DOI: 10.1103/RevModPhys.25.220.
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Chapter 12: Ferromagnetism and Antiferromagnetism

In ferromagnets, neighboring moments align parallel producing spontaneous magnetization below the Curie temperature. In antiferromagnets, they align antiparallel.

12.1 Ferromagnetic Elements

Element	T _C (K)	M _s (0) (kA/m)	μ/μ _B	Structure
Fe	1043	1752	2.22	bcc
Co	1394	1446	1.72	hcp
Ni	631	510	0.60	fcc
Gd	293	2060	7.63	hcp

Non-integer moments (2.22, 1.72, 0.60 μ_B) evidence itinerant ferromagnetism.

12.2 Bloch T^{3/2} Law: Rode & Herrmann (1964)

Rode and Herrmann measured ΔI(T) of Fe, Co, Ni at 4.2-70 K in 20 kOe fields [Rode & Herrmann 1964]. Data extracted directly from the Soviet Physics JETP paper:

Iron ΔI(T) at H = 20 kOe:

T (K)	ΔI (G)	T (K)	ΔI (G)
4.2	0	25.0	0.90
5.4	0.03	27.6	1.05
7.1	0.08	30.4	1.24
9.0	0.15	34.9	1.53
10.0	0.19	40.0	1.84
12.0	0.27	44.8	2.22
13.5	0.34	49.1	2.58
15.9	0.44	53.4	2.95
18.0	0.54	61.0	3.50
21.8	0.76		

From Rode & Herrmann, JETP 19(5), 1081 (1964), Table II.

Bloch law: $\Delta I/I_0 = cT^{3/2}$. Fitted coefficients:

Metal	c times 10 ⁶	Exchange integral A (erg)
Fe	4.5 +/- 0.1	(1.59 pm 0.05) times 10 ⁻¹⁴
Co	2.4 +/- 0.1	(2.92 pm 0.05) times 10 ⁻¹⁴
Ni	9.3 +/- 0.25	(2.3 pm 0.04) times 10 ⁻¹⁴

Nickel deviates from T^{3/2} above 30 K, requiring: $\Delta I/I_0 = cT^{3/2} + dT^{5/2}e^{-\Delta/k_B T}$ with Δ/k_B = 98 pm 4 K -- evidence for a magnon spectrum gap.

12.3 Antiferromagnetic Order: Shull (1951)

Shull, Strauser, and Wollan provided the first experimental proof of AFM order by observing magnetic Bragg peaks in MnO below 120 K [Shull 1951]. This paper has 1321 citations.

Material	T _N (K)	Structure
MnO	116	NaCl (AFM type II)
MnF ₂	67	rutile
FeO	198	NaCl
NiO	525	NaCl
CoO	291	NaCl
Cr	311	bcc (SDW)

12.4 Hysteresis

Material	B _r (T)	H _c (kA/m)	Type
Fe (annealed)	1.3	0.08	very soft
Permalloy	0.6	0.004	very soft
Alnico 5	1.27	50	hard
Nd ₂ Fe ₁₄ B	1.28	880	very hard
SmCo ₅	0.92	720	very hard

Coercive field spans 5 orders of magnitude.

References

1. Rode, V. E. and Herrmann, R. (1964). "Investigation of the Magnetization of Iron, Cobalt, and Nickel at Low Temperatures." JETP 19(5), 1081-1083. -- Fe/Co/Ni $\Delta I(T)$ data at 20 kOe, Bloch $T^{3/2}$ coefficients.
2. Shull, C. G., Strauser, W. A., and Wollan, E. O. (1951). "Neutron Diffraction by Paramagnetic and Antiferromagnetic Substances." Phys. Rev. 83, 333. DOI: 10.1103/PhysRev.83.333. -- 1321 citations.
3. Lynn, J. W. et al. (1984). "Neutron scattering study of the magnon energies in iron at high energy transfers." J. Appl. Phys. 55, 2Sr. DOI: 10.1063/1.333481. -- Fe spin waves to 160 meV.

Chapter 13: Magnetic Resonance

Magnetic resonance techniques exploit the interaction of electromagnetic radiation with magnetic moments in solids. Nuclear magnetic resonance (NMR), electron paramagnetic resonance (EPR), and ferromagnetic resonance (FMR) provide detailed information about local electronic structure, bonding, and magnetic order.

13.1 Nuclear Magnetic Resonance

Discovery

In 1946, Edward Purcell at Harvard and Felix Bloch at Stanford independently detected NMR signals from bulk matter -- Purcell from paraffin wax, Bloch from water. They shared the 1952 Nobel Prize in Physics.

As Alloul (2015, arXiv:1504.06992) describes: "Discovered at the outset of the second world War by F. Bloch and E.M. Purcell, the nuclear magnetic resonance technique has become quite immediately a unique method to investigate the chemical and physical properties of condensed matter."

The Resonance Condition

A nucleus with spin I in a magnetic field B_0 has $2I+1$ energy levels separated by:

[equation]

where γ_n is the gyromagnetic ratio. Resonant absorption occurs at the Larmor frequency:

[equation]

NMR Properties of Common Nuclei

Nucleus	Spin I	$\gamma_n/2\pi$ (MHz/T)	Natural Abundance (%)	Relative Sensitivity
^1H	$1/2$	42.58	99.99	1.000
^{13}C	$1/2$	10.71	1.11	0.016
^{19}F	$1/2$	40.08	100	0.833
^{23}Na	$3/2$	11.27	100	0.093
^{27}Al	$5/2$	11.09	100	0.207
^{29}Si	$1/2$	8.47	4.70	0.008
^{51}V	$7/2$	11.21	99.76	0.383
^{63}Cu	$3/2$	11.29	69.1	0.093

Reference: Alloul, H. (2015). "NMR studies of electronic properties of solids." *Scholarpedia* 9(9):32069. arXiv:1504.06992.

13.2 Knight Shift

In metals, the NMR resonance frequency is shifted from the free-nucleus value by the Knight shift K , arising from the contact

interaction between the nuclear spin and the conduction electron spin density at the nucleus:

[equation]

where A_0 is the contact hyperfine coupling constant and $\chi_{\{P0\}} = (\mu_B)^2 \rho(E_F)/2$ is the Pauli spin susceptibility, directly measuring the electronic density of states at the Fermi level.

The Knight shift is "usually a large quantity which is measured in %. This comes about because the contact coupling A_0 is usually much larger than the corresponding dipole or orbital couplings" (Alloul 2015).

Experimental Knight Shift Values

Metal	K (%)	Metal	K (%)
Li	0.026	Cu	0.237
Na	0.112	Al	0.162
K	0.265	Pt	2.90
Rb	0.653	Sn	0.73

The Knight shift increases with the Pauli susceptibility and the s-electron density at the nucleus.

13.3 Spin-Lattice Relaxation and the Korringa Relation

The spin-lattice relaxation time T_1 measures how quickly the nuclear spin system reaches thermal equilibrium with the lattice. In a simple metal:

[equation]

Combining with the Knight shift yields the Korringa relation:

[equation]

Experimental Verification: ^{27}Al in Aluminum

Alloul (2015) presents data showing that T_1/T of ^{27}Al is constant in pure aluminum over more than three orders of magnitude in temperature:

[equation]

This remarkable constancy validates the Korringa relation and confirms that aluminum is a well-behaved free-electron metal.

Figure description (from arXiv:1504.06992, Fig. 1): "The ^{27}Al spin lattice relaxation time T_1 measured in pure aluminium metal is plotted versus temperature in a log/log scale. The linear fit represents the relation $T_1/T = 1.85 \text{ sec}\cdot\text{K}$."

Deviations from the Korringa Relation

In nearly ferromagnetic metals (e.g., TiBe , Pd), the Stoner enhancement factor $S = 1/(1 - \chi_{\{P0\}})$ modifies the Korringa relation. The Knight shift is more enhanced than $(T_1/T)^{-1}$ because the enhancement is peaked at $\mathbf{q} = 0$.

In nearly antiferromagnetic metals (e.g., MnSi), the opposite occurs: $\chi(\mathbf{q})$ is peaked at the AF wavevector, and the Korringa constant is decreased.

13.4 NMR in Superconductors

Knight Shift Suppression Below T_c

In a BCS superconductor, the pairing of electrons in singlet states suppresses the spin susceptibility. The Knight shift decreases below T_c following the Yosida function:

[equation]

Experimental example (from arXiv:1504.06992, Fig. 2): "The ^{133}Cs and ^{13}C NMR shifts measured in the Cs_2C_{60} phase are plotted versus T below the superconducting temperature $T_c = 30\text{ K}$. The NMR shifts follow the standard Yosida type decrease expected for singlet superconductivity." -- Wzietek et al., Phys. Rev. Lett. 112, 066401 (2014).

Hebel-Slichter Coherence Peak

Below T_c , the spin-lattice relaxation rate first increases above the normal-state Korringa value before decreasing at lower temperatures. This coherence peak (Hebel and Slichter, 1957) results from the enhanced density of states piling up above the superconducting gap.

Experimental example (from arXiv:1504.06992, Fig. 3): "The $\log(1/T_1)$ of ^{119}V in V_2Sn is plotted versus $1/T$ for three distinct applied fields. Below T_c the reduction of T_1 represents the Hebel-Slichter coherence peak. At low T all curves point towards an activated behavior associated with the full opening of the superconducting gap." -- Adapted from MacLaughlin (1976).

13.5 Quadrupole Interactions and NQR

Nuclei with $I > 1/2$ have a non-spherical charge distribution (quadrupole moment Q) that interacts with the electric field gradient (EFG) at the nuclear site. This splits the NMR spectrum into satellite lines.

Nuclear Quadrupole Resonance (NQR)

In zero applied field, the quadrupole interaction alone splits the spin levels. For $I = 3/2$:

[equation]

where ν_Q is the quadrupole frequency and η is the asymmetry parameter.

Experimental example: In Na_2CoO_2 , the ^{23}Na NQR spectrum displays three independent NQR lines corresponding to three distinct Na lattice sites, confirming the ordered Na pattern. -- Platova et al., Phys. Rev. B 80, 224106 (2009).

13.6 Electron Paramagnetic Resonance

EPR (also called ESR) detects unpaired electron spins. The resonance condition is:

[equation]

F Centers in Alkali Halides

Holton et al. (1962) measured the EPR spectra of F centers (electrons trapped at anion vacancies) in KCl. The F center is a single electron in a roughly spherical potential well formed by the surrounding cations.

Experimental observations:

- * Single resonance line near $g = 2.001$ (close to the free electron value)
- * Hyperfine structure from interaction with neighboring nuclear spins (? nearest K neighbors for KCl)
- * Line broadening due to spin-spin interactions between F centers

Reference: Holton, W.C. et al. (1962). "Paramagnetic Resonance of F Centers in Alkali Halides." *Phys. Rev.* 125, 89. (211 citations)

Donor Atoms in Silicon

Feher (1959) performed the classic EPR study of phosphorus donors in silicon, observing:

- * A doublet due to hyperfine interaction with the ^{31}P nucleus ($I = 1/2$)
- * The splitting gives the electron density at the P nucleus: $|\psi(0)|^2$
- * g-factor anisotropy reflecting the ellipsoidal conduction band minima

13.7 Ferromagnetic Resonance

In ferromagnetic resonance (FMR), the entire magnetization of a sample precesses coherently around the applied field. For an ellipsoidal sample:

[equation]

where N_x , N_y , N_z are demagnetization factors.

For a thin film magnetized in-plane: $\omega = \gamma \sqrt{B_0(B_0 + \mu_0 M_s)}$ (Kittel formula).

Spin Wave Resonance

In a thin ferromagnetic film, standing spin waves can be excited, producing additional resonance lines at fields below the uniform FMR mode:

[equation]

where D is the spin-wave stiffness and d is the film thickness.

13.8 RKKY Oscillations Observed by NMR

One of the most striking demonstrations of NMR as a local probe is the direct observation of RKKY spin density oscillations around magnetic impurities in metals.

In dilute Cu-Mn alloys, Alloul (2012) resolved up to 17 distinct shells of Cu neighbors around each Mn impurity, each with a different Knight shift. The spatial dependence follows:

[equation]

Figure description (from arXiv:1504.06992, Fig. 5): "NMR spectra of the ^{63}Cu and ^{65}Cu nuclear spins in dilute Cu-Mn

alloys obtained by sweeping the applied external field, at 1.3 K. On both sides of the central lines one can see the weak NMR signals of the diverse copper shells of neighbors."

Reference: Alloul, H. (2012). "From Friedel oscillations and Kondo effect to the pseudogap in cuprates." *J. Supercond. Nov. Mag.* 25, 385. *arXiv:1204.3804*.

References

1. Alloul, H. (2015). "NMR studies of electronic properties of solids." *Scholarpedia* 9(9):32069. *arXiv:1504.06992* -- Primary source for this chapter
 2. Holton, W.C. et al. (1962). "Paramagnetic Resonance of F Centers in Alkali Halides." *Phys. Rev.* 125, 89.
 3. Hebel, L.C. and Slichter, C.P. (1957). "Nuclear Spin Relaxation in Normal and Superconducting Aluminum." *Phys. Rev.* 113, 1504.
 4. MacLaughlin, D.E. (1976). "Magnetic Resonance in the Superconducting State." *Solid State Physics* 31, 1.
 5. Wzietek, P. et al. (2014). "NMR study of the Superconducting gap variation near the Mott transition in Cs_2CoO_2 ." *Phys. Rev. Lett.* 112, 066401.
 6. Alloul, H. (2012). "From Friedel oscillations and Kondo effect to the pseudogap in cuprates." *J. Supercond. Nov. Mag.* 25, 385. *arXiv:1204.3804*.
 7. Platova, T.A. et al. (2009). "NQR and X-ray investigation of the structure of $\text{Na}_{1-x}\text{CoO}_2$." *Phys. Rev. B* 80, 224106.
 8. Abragam, A. (1961). *The Principles of Nuclear Magnetism*. Oxford: Clarendon Press.
 9. Slichter, C.P. (1963). *Principles of Magnetic Resonance*. Harper and Row (3rd ed. 1989, Springer).
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Part VI: Optical and Dielectric Properties

Chapter 14: Plasmons, Polaritons, and Polarons

Bulk Plasmon Energy Loss in Metals

Electron energy-loss spectroscopy (EELS) provides direct measurement of bulk plasmon energies. Powell and Swan (1959) transmitted keV electrons through thin Al films and observed a dominant loss peak at 15.3 eV, confirming the free-electron plasmon prediction.

"The energy losses observed in aluminium are 15.0 +/- 0.3 eV... in good agreement with the value 15.7 eV calculated from the free-electron formula." -- Powell & Swan, Phys. Rev. 115, 869 (1959)

Metal	Measured plasmon energy (eV)	Free-electron prediction (eV)	Electron density n (10^{28} m^{-3})
Al	15.3	15.8	18.1
Mg	10.6	10.9	8.61
Si	16.5	16.0	--
Na	5.7	5.9	2.65
K	3.7	4.3	1.40
Be	18.7	18.4	24.7

The close agreement between measured and free-electron values for simple metals validates the Drude dielectric function. Deviations in Si reflect interband transition contributions.

Surface Plasmons

Ritchie (1957) predicted surface plasmon modes at energy $\omega_s = \omega_p / \sqrt{2}$. For Al, the surface plasmon appears at 10.3 eV in reflection EELS. Surface plasmons are now routinely excited by infrared light on nanostructured metal surfaces (localized surface plasmon resonance), forming the basis of plasmonic biosensors.

Lyddane-Sachs-Teller Relation and Polaritons

The LST relation connects the static and high-frequency dielectric constants to the longitudinal and transverse optical phonon frequencies: $\epsilon(0)/\epsilon(\infty) = (\omega_{LO} / \omega_{TO})^2$.

Crystal	$\omega_{LO} \text{ (cm}^{-1}\text{)}$	$\omega_{TO} \text{ (cm}^{-1}\text{)}$	$\epsilon(0)/\epsilon(\infty) \text{ measured}$	$(\omega_{LO}/\omega_{TO})^2$
GaAs	292	269	1.18	1.18
InSb	197	185	1.14	1.13
NaCl	264	164	5.9	2.59
LiF	659	307	8.9	4.61

For strongly ionic crystals (NaCl, LiF) the LO-TO splitting is large, producing a wide reststrahlen band where infrared reflectivity approaches unity.

Polariton Dispersion

Henry and Hopfield (1965) measured the polariton dispersion in GaP using Raman scattering, demonstrating the anticrossing between photon and TO-phonon branches. The phonon-polariton gap between ω_{TO} and ω_{LO} is directly observable as a forbidden frequency region in infrared transmission.

Polarons and the Mott Metal-Insulator Transition

The electron-phonon coupling in polar semiconductors dresses carriers as polarons. A dramatic consequence of electron-electron and electron-lattice interactions is the Mott transition. Rosenbaum et al. (1983) studied the metal-insulator transition in Si:P, finding the critical phosphorus concentration $n_c = 3.74 \times 10^{18} \text{ cm}^{-3}$.

"We find the transition to be continuous... the critical concentration $n_c = 3.74 \pm 0.04 \times 10^{18} \text{ cm}^{-3}$." -- Rosenbaum et al., Phys. Rev. B 27, 7509 (1983)

Dopant system	Critical concentration $n_c \text{ (cm}^{-3}\text{)}$	Mott criterion $(n_c)^{1/3} a_B$
Si:P	3.74×10^{18}	0.26
Si:B	4.06×10^{18}	0.31
Si:As	8.5×10^{18}	0.25
Ge:Sb	1.2×10^{17}	0.26

The Mott criterion $(n_c)^{1/3} a_B \sim 0.26$ holds remarkably well across different host-dopant combinations.

Polaron Effective Mass

Material	α (Froehlich coupling)	m^*/m_{band}	Method
InSb	0.014	1.01	Cyclotron res.
GaAs	0.068	1.03	Cyclotron res.
CdTe	0.39	1.05	Magneto-optical
AgBr	1.53	1.26	Optical abs.
SrTiO ₃	3.77	~ 3	Transport
KCl	3.44	2.5	Optical abs.

Weak-coupling materials ($\alpha < 1$) show small mass enhancement; strong-coupling materials approach small-polaron behavior.

References

1. C. J. Powell and J. B. Swan, "Origin of the Characteristic Electron Energy Losses in Aluminum," Phys. Rev. 115, 869 (1959). DOI: 10.1103/PhysRev.115.869
2. T. F. Rosenbaum et al., "Metal-insulator transition in a doped semiconductor," Phys. Rev. B 27, 7509 (1983). DOI: 10.1103/PhysRevB.27.7509
3. R. H. Ritchie, "Plasma Losses by Fast Electrons in Thin Films," Phys. Rev. 106, 874 (1957). DOI: 10.1103/PhysRev.106.874
4. C. H. Henry and J. J. Hopfield, "Raman Scattering by Polaritons," Phys. Rev. Lett. 15, 964 (1965). DOI: 10.1103/PhysRevLett.15.964
5. R. P. Feynman, "Slow Electrons in a Polar Crystal," Phys. Rev. 97, 660 (1955). DOI: 10.1103/PhysRev.97.660

Chapter 15: Optical Processes and Excitons

Dielectric Function of Semiconductors

The complex dielectric function $\epsilon(\omega) = \epsilon_1 + i\epsilon_2$ encodes all linear optical response. Aspnes and Studna (1983) performed spectroscopic ellipsometry on Si and Ge, providing the standard reference data used in semiconductor optics.

"We report pseudodielectric function spectra $\langle \epsilon(\omega) \rangle$ for Si, Ge, GaP, GaAs, GaSb, InP, InAs, and InSb from 1.5 to 6.0 eV." -- Aspnes & Studna, Phys. Rev. B 27, 985 (1983)

Material	E_1 critical point (eV)	E_2 critical point (eV)	ϵ_1 at 2 eV	ϵ_2 at 4.3 eV
Si	3.4	4.25	15.3	38.5
Ge	2.1	4.35	21.6	29.7
GaAs	2.9	4.7	12.0	26.8
GaP	3.7	5.3	10.2	18.1
InP	3.2	4.7	11.4	20.0

The critical-point structure in ϵ_2 directly maps the joint density of states at van Hove singularities in the band structure.

Raman Scattering in Diamond

Raman spectroscopy probes zone-center optical phonons. Diamond has a single first-order Raman-active mode (T_{2g} symmetry). Liu et al. (2000) studied the diamond Raman line under high pressure.

"The zone-center optical phonon frequency of diamond at ambient conditions is 1332.5 cm⁻¹." -- Liu et al., Phys. Rev. B 61, 3391 (2000)

Material	Raman frequency (cm ⁻¹)	Symmetry	Linewidth (cm ⁻¹)
Diamond	1332.5	T _{2g}	1.2
Si	520.7	T _{2g}	3.5
Ge	300.7	T _{2g}	3.2
GaAs (LO)	292	--	3.0
GaAs (TO)	269	--	2.5
6H-SiC (A ₁)	967	A ₁ (LO)	2.8

The narrow linewidth of diamond (1.2 cm⁻¹) reflects its exceptionally long phonon lifetime due to limited anharmonic decay channels.

Excitons: Wannier and Frenkel Types

Excitons are bound electron-hole pairs. In semiconductors with large dielectric constants, the Wannier exciton binding energy follows a hydrogen-like series: $E_n = -E_x / n^2$.

Material	Exciton binding energy E_x (meV)	Bohr radius a_x (nm)	Type
GaAs	4.2	13	Wannier

CdS	28	2.8	Wannier
ZnO	60	1.8	Wannier
Cu ₂ O	97	1.1	Wannier
NaCl	~1000	~0.3	Frenkel
Anthracene	~1000	~0.5	Frenkel

Giant Rydberg Excitons in Cu₂O

Kazimierczuk et al. (2014) observed Rydberg exciton states in Cu₂O up to principal quantum number $n = 25$, a landmark achievement in semiconductor spectroscopy.

"Here we observe giant Rydberg excitons with principal quantum numbers as large as $n = 25$ in cuprous oxide... the coherence of these exciton states is remarkable." -- Kazimierczuk et al., Nature 514, 343 (2014)

Selected absorption peak positions from the yellow exciton series in Cu₂O at 1.2 K:

n	Energy (eV)	Linewidth (micro-eV)
2	2.1478	--
5	2.1680	150
10	2.1710	40
15	2.1718	20
20	2.1721	12
25	2.1723	8

The series converges to the band gap at 2.17208 eV. The exciton size at $n = 25$ exceeds 1 micrometer, making it a mesoscopic quantum object. Rydberg blockade effects were observed at high densities.

Optical Absorption Edge

The absorption coefficient α near the band edge distinguishes direct and indirect gap semiconductors:

* Direct gap: $\alpha \sim (h\nu - E_g)^{1/2}$

* Indirect gap: $\alpha \sim (h\nu - E_g \pm E_{\text{phonon}})^2$

GaAs	1.42	Direct	~8000
InP	1.34	Direct	~5000
Si	1.12	Indirect	~50
Ge	0.66	Indirect	~300
GaN	3.39	Direct	~10 ⁵

The sharp onset in direct-gap materials makes them ideal for optoelectronic devices.

References

1. D. E. Aspnes and A. A. Studna, "Dielectric functions and optical parameters of Si, Ge, GaP, GaAs, GaSb, InP, InAs, and InSb from 1.5 to 6.0 eV," Phys. Rev. B 27, 985 (1983). DOI: 10.1103/PhysRevB.27.985

2. T. Kazimierzczuk et al., "Giant Rydberg excitons in the copper oxide Cu₂O," *Nature* 514, 343 (2014). DOI: 10.1038/nature13832
 3. L. Liu et al., "High-pressure Raman study of diamond," *Phys. Rev. B* 61, 3391 (2000). DOI: 10.1103/PhysRevB.61.3391
 4. R. J. Elliott, "Intensity of Optical Absorption by Excitons," *Phys. Rev.* 108, 1384 (1957). DOI: 10.1103/PhysRev.108.1384
 5. P. Y. Yu and M. Cardona, *Fundamentals of Semiconductors*, 4th ed. (Springer, 2010).
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Chapter 16: Dielectrics and Ferroelectrics

Dielectric Constants of Insulators

The dielectric constant ϵ_r describes the polarization response of a material to an applied electric field. Static values at room temperature:

Material	ϵ_r (static)	ϵ_r (optical)	Dominant mechanism
Vacuum	1.000	1.000	--
Diamond	5.7	5.7	Electronic
NaCl	5.9	2.34	Ionic + electronic
SiO ₂ (quartz)	3.78	2.13	Ionic + electronic
H ₂ O (liquid)	80.4	1.78	Orientational
BaTiO ₃ (25 C)	~1700	5.4	Ferroelectric
SrTiO ₃ (4 K)	~24000	5.2	Quantum paraelectric

Ferroelectric Phase Transition in BaTiO₃

Barium titanate is the prototypical displacive ferroelectric. Samara (1971) measured the temperature dependence of the dielectric constant through the cubic-to-tetragonal transition at $T_C = 120$ C.

"At the Curie point, the dielectric constant shows a sharp peak exceeding 10,000, consistent with a first-order transition." -- Samara, Ferroelectrics 2, 277 (1971)

Temperature (C)	ϵ_r (heating)	Phase
25	1700	Tetragonal
80	3200	Tetragonal
100	5500	Tetragonal
115	9000	Tetragonal
120 (T_C)	>10000	Transition
130	6200	Cubic
150	3500	Cubic
200	2000	Cubic

Above T_C , the Curie-Weiss law applies: $\epsilon_r = C / (T - T_0)$ with $C \sim 1.5 \times 10^5$ K and $T_0 \sim 383$ K.

BaTiO₃ Structural Phase Sequence

Phase	Temperature range	Crystal system	Spontaneous polarization (micro-C/cm ²)
Rhombohedral	< -90 C	Rhombohedral	~10 along [111]
Orthorhombic	-90 to 5 C	Orthorhombic	~14 along [011]
Tetragonal	5 to 120 C	Tetragonal	26 along [001]
Cubic	> 120 C	Cubic (Pm3m)	0

Polarization-Electric Field Hysteresis

The P-E hysteresis loop is the defining signature of ferroelectricity. For BaTiO₃ single crystal at room temperature:

- * Saturation polarization P_s : 26 micro-C/cm²
- * Remanent polarization P_r : 22 micro-C/cm²
- * Coercive field E_c : ~1 kV/cm

For comparison, PZT (PbZr_{0.52}Ti_{0.48}O₃) ceramics show $P_r \sim 35$ micro-C/cm² and $E_c \sim 10$ -15 kV/cm.

Piezoelectric Coefficients

Piezoelectric materials convert between mechanical stress and electric polarization. The d_{33} coefficient (longitudinal) is a key figure of merit.

Material	d_{33} (pC/N)	T_C (C)	Application
Quartz (alpha-SiO ₂)	2.3	573	Frequency standards
BaTiO ₃ (ceramic)	190	120	Capacitors
PZT-5A	374	365	Actuators, transducers
PZT-5H	593	193	Sensors
PMN-PT single crystal	1500-2800	130-170	Medical ultrasound
PVDF polymer	-33	--	Flexible sensors
LiNbO ₃	6	1210	SAW devices, optics
AlN	5.5	--	MEMS, thin film BAW

Soft Mode Theory

The ferroelectric transition is driven by the softening of a transverse optical phonon. As T approaches T_C from above, the soft mode frequency decreases: $\omega_{TO}^2 \sim (T - T_C)$.

Cowley (1962) and Shirane et al. (1970) confirmed soft mode behavior in SrTiO₃ and BaTiO₃ via neutron scattering. In SrTiO₃, the soft mode frequency drops to ~ 10 cm⁻¹ at 4 K but never reaches zero -- SrTiO₃ is a quantum paraelectric where zero-point fluctuations suppress long-range order.

Material	Soft mode frequency at $T_C + 50$ K (cm ⁻¹)	Transition type
BaTiO ₃	~50	First order
PbTiO ₃	~80	First order
SrTiO ₃	~12 (at 4 K)	Suppressed
KNbO ₃	~55	First order

References

1. G. A. Samara, "The Effect of Pressure on the Dielectric Properties of BaTiO₃," *Ferroelectrics* 2, 277 (1971). DOI: 10.1080/00150197108234102
2. G. Shirane et al., "Soft Ferroelectric Modes in Lead Titanate," *Phys. Rev. B* 2, 155 (1970). DOI: 10.1103/PhysRevB.2.155
3. R. A. Cowley, "Temperature Dependence of a Transverse Optic Mode in Strontium Titanate," *Phys. Rev. Lett.* 9, 159

(1962). DOI: 10.1103/PhysRevLett.9.159

4. S.-E. Park and T. R. Shrout, "Ultrahigh strain and piezoelectric behavior in relaxor based ferroelectric single crystals," J. Appl. Phys. 82, 1804 (1997). DOI: 10.1063/1.365983

5. M. E. Lines and A. M. Glass, Principles and Applications of Ferroelectrics and Related Materials (Oxford, 1977).

Part VII: Surfaces, Nanostructures, and Defects

Chapter 17: Surface and Interface Physics

Work Functions of Metals

The work function ϕ is the minimum energy to remove an electron from the metal surface to vacuum. Michaelson (1977) compiled a comprehensive review of recommended values.

"A compilation of recommended values of work functions for polycrystalline and single crystal surfaces of the elements is presented." -- Michaelson, J. Appl. Phys. 48, 4729 (1977)

Metal	ϕ (eV), polycrystalline	ϕ (eV), (111) face	Electron config.
Cs	2.14	1.95	[Xe]6s ¹
K	2.30	2.30	[Ar]4s ¹
Na	2.75	--	[Ne]3s ¹
Al	4.28	4.24	[Ne]3s ² 3p ¹
Cu	4.65	4.94	[Ar]3d ¹⁰ 4s ¹
Au	5.10	5.31	[Xe]4f ¹⁴ 5d ¹⁰ 6s ¹
Pt	5.65	5.93	[Xe]4f ¹⁴ 5d ⁹ 6s ¹
W	4.55	4.47	[Xe]4f ¹⁴ 5d ⁴ 6s ²

Low work-function metals (Cs, K) are used as photocathodes; high work-function metals (Pt, Au) are preferred for Schottky barrier contacts.

Surface Reconstruction: Si(111) 7x7

Clean semiconductor surfaces reconstruct to minimize dangling-bond energy. The Si(111)7x7 reconstruction, solved by Takayanagi et al. (1985) using transmission electron diffraction, contains 19 dangling bonds per unit cell (reduced from 49 in the unreconstructed surface). STM images by Binnig et al. (1983) provided the first real-space visualization.

p-n Junction Current-Voltage Characteristics

The Shockley diode equation describes the ideal p-n junction:

$$I = I_0 [\exp(eV / nkT) - 1]$$

Parameter	Si diode (typical)	Ge diode (typical)	GaAs diode
Saturation current I_0	$\sim 10^{-12}$ A	$\sim 10^{-7}$ A	$\sim 10^{-18}$ A
Ideality factor n	1.0-2.0	1.0-1.5	1.0-2.0
Built-in voltage V_{bi}	0.7 V	0.3 V	1.1 V
Breakdown voltage	10-1000 V	5-200 V	5-100 V

LED Emission Wavelengths

Light-emitting diodes exploit direct-gap semiconductor junctions. The emission wavelength is set by the band gap.

Material system	Band gap (eV)	Wavelength (nm)	Color	Year demonstrated
AlGaInP	1.9-2.3	540-650	Green-Red	1990s
GaAsP	1.8-2.1	590-690	Yellow-Red	1962
GaAs	1.42	870	Infrared	1962
InGaN	2.0-3.4	365-520	UV-Green	1993
InGaN (blue)	2.64	470	Blue	1993 (Nakamura)
InGaN (white)	--	broadband	White (phosphor)	1996
AlGaN	3.4-6.2	200-365	Deep UV	2000s

Nakamura's development of efficient InGaN blue LEDs (Nobel Prize 2014) enabled white solid-state lighting.

Integer Quantum Hall Effect

Von Klitzing, Dorda, and Pepper (1980) discovered the quantization of the Hall resistance in a Si MOSFET 2DEG at low temperature and high magnetic field.

"The Hall voltage of a silicon MOS inversion layer shows well-defined plateaus... $R_H = h/ie^2$, where i is an integer." -- von Klitzing et al., PRL 45, 494 (1980)

Plateau index i	$R_H = h/(ie^2)$ (Ohm)	Measured accuracy
1	25812.807	parts per 10^9
2	12906.403	parts per 10^9
3	8604.269	parts per 10^9
4	6453.202	parts per 10^{10}

The von Klitzing constant $R_K = h/e^2 = 25812.80745...$ Ohm is now used as the resistance standard.

Fractional Quantum Hall Effect

Tsui, Stormer, and Gossard (1982) discovered fractional quantization in high-mobility GaAs/AlGaAs heterostructures.

"We observe a quantized Hall plateau... and a minimum in the diagonal resistivity... at filling factor $\nu = 1/3$." -- Tsui et al., PRL 48, 1559 (1982)

Fraction ν	R_H (h/e^2)	Discovery year	Explanation
1/3	$3 h/e^2$	1982	Laughlin wavefunction
2/3	$3/2 h/e^2$	1982	Particle-hole of 1/3
2/5	$5/2 h/e^2$	1983	Composite fermions
5/2	$2/5 h/e^2$	1987	Non-Abelian (proposed)

The FQHE states arise from strong electron-electron interactions forming an incompressible quantum liquid.

References

1. H. B. Michaelson, "The work function of the elements and its periodicity," J. Appl. Phys. 48, 4729 (1977). DOI: 10.1063/1.323539
2. K. von Klitzing, G. Dorda, and M. Pepper, "New Method for High-Accuracy Determination of the Fine-Structure

Constant Based on Quantized Hall Resistance," Phys. Rev. Lett. 45, 494 (1980). DOI: 10.1103/PhysRevLett.45.494

3. D. C. Tsui, H. L. Stormer, and A. C. Gossard, "Two-Dimensional Magnetotransport in the Extreme Quantum Limit," Phys. Rev. Lett. 48, 1559 (1982). DOI: 10.1103/PhysRevLett.48.1559

4. S. Nakamura et al., "High-brightness InGaN blue, green and yellow light-emitting diodes with quantum well structures," Jpn. J. Appl. Phys. 34, L797 (1995). DOI: 10.1143/JJAP.34.L797

5. G. Binnig et al., "7x7 Reconstruction on Si(111) Resolved in Real Space," Phys. Rev. Lett. 50, 120 (1983). DOI: 10.1103/PhysRevLett.50.120

Chapter 18: Nanostructures

When the dimensions of a solid are reduced to the nanometer scale -- comparable to the electron wavelength or the mean free path -- quantum confinement and surface effects dominate. This chapter presents experimental data on imaging, manipulation, and electronic transport in one-dimensional (1D) and zero-dimensional (0D) systems.

18.1 Imaging Techniques for Nanostructures

Scanning Tunneling Microscopy

The scanning tunneling microscope (STM), invented by Binnig and Rohrer in 1981 (Nobel Prize 1986), images surfaces by measuring the quantum mechanical tunneling current between a sharp metallic tip and the sample surface. The tunneling current depends exponentially on the tip-sample distance:

[equation]

where ϕ is the work function ($\sim 4\text{--}5$ eV for metals) and d is the gap. A change of just $1 \text{ \AA}$ in d changes the current by an order of magnitude -- this is the basis for atomic resolution.

Atomic Manipulation with the STM

The landmark experiment: Eigler and Schweizer (1990). Don Eigler and Erhard Schweizer at IBM Almaden used the STM at 4 K to position 35 individual xenon atoms on a Ni(110) surface, spelling out "IBM." This was published in Nature (2708 citations) and remains one of the most iconic images in physics.

"Here we report the use of the STM at low temperatures (4 K) to position individual xenon atoms on a single-crystal nickel surface with atomic precision. This capacity has allowed us to fabricate rudimentary structures of our own design, atom by atom." -- Eigler and Schweizer, Nature 344, 524 (1990).

The technique: The STM tip is brought close enough to the Xe atom ($\sim 4 \text{ \AA}$) that the van der Waals attraction between tip and atom exceeds the lateral diffusion barrier on the surface. The atom then follows the tip as it is moved laterally. At the desired position, the tip is retracted to release the atom.

The atomic switch (1991). Eigler, Lutz, and Rudge demonstrated that a single Xe atom could be transferred reversibly between the STM tip and the Ni surface by voltage pulses -- the first single-atom switch. Published in Nature 352, 600 (1991) with 1132 citations.

"We report the use of the STM to transfer a Xe atom back and forth between the tip of the microscope and a nickel surface." -- Eigler et al., Nature 352, 600 (1991).

Quantum corrals (1993). Crommie, Lutz, and Eigler arranged 48 iron atoms in a circular corral (radius $\sim 71 \text{ \AA}$) on Cu(111), directly imaging the standing wave pattern of surface-state electrons confined inside. This provided a stunning real-space visualization of quantum confinement.

Modern Developments

Morgenstern, Lorente, and Rieder (2013) reviewed the evolution of atomic manipulation from individual atom positioning to scalable fabrication. Recent advances include:

- * Automated atomic assembly using deep reinforcement learning (2022, 33 citations in Nature Communications)
 - * Coherent spin manipulation of individual atoms using pulsed STM (2019, Science, 161 citations)
 - * Creating designer quantum states atom-by-atom, including topological and magnetic phases (2019, Nature Reviews Physics, 142 citations)
-

18.2 Conductance Quantization

Theory

In a quantum point contact -- a narrow constriction between two electron reservoirs -- the electrical conductance is quantized in units of the conductance quantum:

[equation]

Each transverse mode that fits through the constriction contributes one quantum of conductance. As the constriction is widened (by gate voltage), the conductance increases in steps of G_0 .

Experimental Discovery

This was first observed independently by:

- * van Wees et al. (1988) in a GaAs/AlGaAs quantum point contact defined by split gates
- * Wharam et al. (1988) in a similar split-gate device

Both groups observed clean conductance staircases with plateaus at integer multiples of G_0 , confirming the Landauer formula:

[equation]

where T_n is the transmission probability of mode n .

Conductance Quantization in Metallic Contacts

Gao et al. (2025) demonstrated conductance quantization in Cu quantum point contacts fabricated using STM atomic manipulation, showing that the quantization persists even in metallic systems with much shorter Fermi wavelengths.

18.3 Quantum Dots: Artificial Atoms

A quantum dot confines electrons in all three dimensions, producing discrete energy levels like an atom.

Coulomb Blockade

When the charging energy $E_C = e^2/(2C)$ exceeds $k_B T$, adding a single electron to the dot requires a measurable voltage step $\Delta V = e/C$. As the gate voltage is swept, current flows only at discrete values -- Coulomb oscillations.

Characteristic Energy Scales

Dot Type	Charging Energy E_C (meV)	Level Spacing ΔE (meV)
GaAs lateral dot ($d \sim 100$ nm)	1-5	0.1-0.5

Self-assembled InAs dot	20-50	10-30
Metallic nanoparticle (d sim 5 nm)	50-200	0.01-0.1
Single molecule (C??)	200-500	50-100

Allerbeck et al. (2025, Nature Communications) demonstrated ultrafast Coulomb blockade in an atomic-scale quantum dot, resolving charge transport through quantized defect states on femtosecond timescales.

Semiconductor Nanocrystals (Quantum Dots)

Colloidal semiconductor nanocrystals (CdSe, InP, PbS) exhibit size-tunable photoluminescence: as the dot diameter decreases from ~6 nm to ~2 nm, the emission shifts from red to blue due to quantum confinement.

The confinement energy for a spherical dot of radius R:

[equation]

This work, pioneered by Louis Brus and Moungi Bawendi, was recognized with the 2023 Nobel Prize in Chemistry.

The Kondo Effect in Quantum Dots

When a quantum dot has an odd number of electrons, it behaves like a magnetic impurity coupled to leads -- a nanoscale realization of the Kondo effect (Ch. 22). Below the Kondo temperature T_K , the dot's spin is screened by the conduction electrons, and the conductance reaches the unitary limit $G = 2e^2/h$.

18.4 Vibrational and Thermal Properties of Nanostructures

Quantized Vibrational Modes

In nanostructures, the phonon spectrum is modified by confinement. For a nanoparticle of diameter d, the lowest vibrational mode has frequency:

[equation]

where v_s is the sound velocity. For $d = 5$ nm and $v_s = 5000$ m/s: $\nu \sim 1$ THz.

Heat Capacity

The heat capacity of nanostructures deviates from the bulk Debye T^3 law at low temperatures because the phonon spectrum has a minimum frequency cutoff set by the particle size.

References

1. Eigler, D.M. and Schweizer, E.K. (1990). "Positioning single atoms with a scanning tunnelling microscope." Nature 344, 524. (2708 citations)
2. Eigler, D.M., Lutz, C.P., and Rudge, W.E. (1991). "An atomic switch realized with the scanning tunnelling microscope." Nature 352, 600. (1132 citations)
3. Crommie, M.F., Lutz, C.P., and Eigler, D.M. (1993). "Confinement of electrons to quantum corrals on a metal surface."

Science 262, 218.

4. van Wees, B.J. et al. (1988). "Quantized conductance of point contacts in a two-dimensional electron gas." Phys. Rev. Lett. 60, 848.

5. Wharam, D.A. et al. (1988). "One-dimensional transport and the quantisation of the ballistic resistance." J. Phys. C 21, L209.

6. Morgenstern, K., Lorente, N., and Rieder, K.-H. (2013). "Controlled manipulation of single atoms and small molecules using the scanning tunnelling microscope." Phys. Status Solidi B 250, 1671.

7. Allerbeck, J. et al. (2025). "Ultrafast Coulomb blockade in an atomic-scale quantum dot." Nature Comm.

8. Yang, J. et al. (2019). "Coherent spin manipulation of individual atoms on a surface." Science 366, 509. (161 citations)

9. Khajetoorians, A.A. et al. (2019). "Creating designer quantum states of matter atom-by-atom." Nature Reviews Physics 1, 703. (142 citations)

Chapter 19: Noncrystalline Solids

Short-Range Order: Radial Distribution Functions

Amorphous solids lack long-range periodical order but retain short-range order characterized by the radial distribution function (RDF) $g(r)$. Laaziri et al. (1999) performed high-resolution X-ray diffraction on amorphous silicon, providing the most precise RDF measurement.

"The first-neighbor distance in a-Si is 2.35 +/- 0.01 Å with a coordination number of 3.88 +/- 0.05, very close to the crystalline value of 4." -- Laaziri et al., Phys. Rev. B 60, 13520 (1999)

Parameter	a-Si (Laaziri 1999)	c-Si (diamond)	Difference
First-neighbor distance	2.35 Å	2.35 Å	< 0.01 Å
Coordination number	3.88	4.00	-3%
Bond angle	109.5 +/- 10 deg	109.47 deg	~10 deg spread
Second-neighbor distance	3.83 Å	3.84 Å	~0.01 Å
Second-neighbor width	0.25 Å (FWHM)	< 0.05 Å	broadened

The key finding: bond lengths are preserved to high accuracy; the disorder manifests primarily as bond-angle distortions.

Vitreous SiO₂ Structure

Wright (1994) compiled neutron and X-ray diffraction data for vitreous silica, the archetypal network glass.

Structural parameter	Vitreous SiO ₂	alpha-Quartz
Si-O bond length	1.608 Å	1.609, 1.612 Å
O-Si-O angle	109.5 deg	108.8-110.5 deg
Si-O-Si angle	144 +/- 35 deg	143.7 deg
Si coordination	4.0	4
O coordination	2.0	2
Density	2.20 g/cm ³	2.65 g/cm ³

Vitreous SiO₂ consists of corner-sharing SiO₄ tetrahedra with a broad distribution of Si-O-Si bridging angles, producing a continuous random network (Zachariasen model).

Glass Transition Temperatures

The glass transition temperature T_g marks the crossover from supercooled liquid to glassy solid. It depends on cooling rate but is reproducible for standard rates (~10 K/min).

Material	T_g (K)	T_m (K)	T_g / T_m	Type
SiO ₂	1473	1996	0.74	Oxide network
B ₂ O ₃	530	723	0.73	Oxide network
GeO ₂	815	1389	0.59	Oxide network
Na ₂ O-2SiO ₂	727	1147	0.63	Modified oxide

Polystyrene	373	~513	0.73	Polymer
Glycerol	190	291	0.65	Molecular
Se	310	494	0.63	Chalcogenide
As ₂ S ₃	478	585	0.82	Chalcogenide
Zr ₄₁ Ti ₁₄ Cu _{12.5} Ni ₁₀ Be _{22.5}	623	993	0.63	Metallic (Vit1)
Pd ₄₀ Ni ₄₀ P ₂₀	590	884	0.67	Metallic

The ratio $T_g/T_m \sim 2/3$ (Kauzmann's rule) holds approximately for many glass formers.

Electronic Properties of Amorphous Semiconductors

Anderson localization in disordered systems creates a mobility edge separating extended and localized states. The density of states in a-Si:H features:

Energy region	States	Density (cm ⁻³ eV ⁻¹)
Conduction band (extended)	Extended above E_c	$\sim 10^{21}$
Conduction band tail	Localized, exponential	$\sim 10^{18}$ - 10^{20}
Mid-gap defects (dangling bonds)	Localized	$\sim 10^{15}$ - 10^{16} (a-Si:H)
Valence band tail	Localized, exponential	$\sim 10^{18}$ - 10^{20}
Valence band (extended)	Extended below E_v	$\sim 10^{21}$

Hydrogenation (a-Si:H) reduces the dangling-bond density from $\sim 10^{19}$ cm⁻³ in pure a-Si to $\sim 10^{15}$ - 10^{16} cm⁻³, enabling device applications (solar cells, TFTs).

Kauzmann Paradox and Fragility

Angell (1995) classified glass formers by their fragility -- the departure from Arrhenius viscosity behavior.

Glass former	Fragility index m	Classification
SiO ₂	20	Strong
GeO ₂	20	Strong
Glycerol	53	Intermediate
o-Terphenyl	81	Fragile
PVC	191	Very fragile

Strong liquids (SiO₂) show Arrhenius viscosity; fragile liquids show dramatic super-Arrhenius slowdown near T_g .

References

1. K. Laaziri et al., "High-resolution radial distribution function of pure amorphous silicon," Phys. Rev. B 60, 13520 (1999). DOI: 10.1103/PhysRevB.60.13520
2. A. C. Wright, "Neutron scattering from vitreous silica. V. The structure of vitreous silica: What have we learned from 60 years of diffraction studies?," J. Non-Cryst. Solids 179, 84 (1994). DOI: 10.1016/0022-3093(94)90687-4
3. C. A. Angell, "Formation of Glasses from Liquids and Biopolymers," Science 267, 1924 (1995). DOI: 10.1126/science.267.5206.1924
4. P. W. Anderson, "Absence of Diffusion in Certain Random Lattices," Phys. Rev. 109, 1492 (1958). DOI:

10.1103/PhysRev.109.1492

5. R. A. Street, Hydrogenated Amorphous Silicon (Cambridge University Press, 1991).

Chapter 20: Point Defects

Vacancy Formation Energy

The equilibrium vacancy concentration follows $n_v / N = \exp(-E_f / kT)$, where E_f is the formation energy. Simmons and Balluffi (1960) determined E_f in aluminum by simultaneously measuring thermal expansion (all sites, including vacant) and lattice parameter (occupied sites only).

"The equilibrium concentration of vacancies in aluminum near the melting point is approximately 9.4×10^{-4} , corresponding to a formation energy of 0.76 eV." -- Simmons & Balluffi, Phys. Rev. 117, 52 (1960)

Metal	E_f (eV)	E_m (migration, eV)	$E_a = E_f + E_m$ (eV)	Method
Al	0.76	0.65	1.41	Differential dilatometry
Cu	1.28	0.70	1.98	Quenching + resistivity
Au	0.94	0.83	1.77	Quenching + resistivity
Ag	1.09	0.66	1.75	Positron annihilation
Pt	1.35	1.43	2.78	Quenching experiments
W	3.60	1.70	5.30	Resistivity recovery
Fe	1.60	0.55	2.15	Resistivity recovery
Ni	1.79	1.04	2.83	Positron annihilation

High-melting-point metals (W, Pt) have correspondingly large formation energies. The vacancy concentration in Al at the melting point (933 K) is $\sim 10^{-4}$, meaning roughly 1 in 10,000 sites is vacant.

F-Center Absorption in Alkali Halides

F-centers (Farbe = color) are electrons trapped at anion vacancies. They produce broad optical absorption bands that give alkali halide crystals their characteristic colors. Seitz (1946) provided the first comprehensive review.

"The color centers in alkali halides provide a beautiful example of the quantum mechanics of a particle in a potential well." -- Seitz, Rev. Mod. Phys. 18, 384 (1946)

Crystal	F-center absorption peak (nm)	F-center peak (eV)	Lattice constant a (Å)	Color of crystal
LiF	248	5.00	4.03	Colorless (UV)
LiCl	385	3.22	5.13	Yellow
NaF	340	3.65	4.63	Pale yellow
NaCl	465	2.67	5.64	Yellow-brown
NaBr	540	2.30	5.97	Brown
KCl	563	2.20	6.29	Violet
KBr	620	2.00	6.60	Blue-green
RbCl	620	2.00	6.58	Blue-green

Mollwo-Ivey Relation

The F-center absorption energy scales with the lattice constant as:

$$E_F = A / a^n$$

where $A \sim 17.7 \text{ eV} \cdot \text{\AA}^n$ and $n \sim 1.84$. This empirical relation follows from treating the F-center as a particle in a box of size proportional to the lattice constant.

Crystal pair	a_1 / a_2	E_1 / E_2 (measured)	E_1 / E_2 (predicted, $n=1.84$)
LiF / NaCl	0.714	1.87	1.86
NaCl / KCl	0.897	1.21	1.20
KCl / KBr	0.953	1.10	1.09
NaBr / KBr	0.905	1.15	1.18

The excellent agreement confirms the simple particle-in-a-box picture.

Schottky and Frenkel Defect Equilibria

Crystal	Dominant defect type	Formation energy (eV/pair)	Method
NaCl	Schottky	2.32	Ionic conductivity
KCl	Schottky	2.50	Ionic conductivity
KBr	Schottky	2.53	Ionic conductivity
AgBr	Frenkel (cation)	1.10	Ionic conductivity
AgCl	Frenkel (cation)	1.45	Ionic conductivity
CaF ₂	Frenkel (anion)	2.70	Ionic conductivity
UO ₂	Frenkel (anion)	3.00	Neutron diffraction

Positron Annihilation Spectroscopy

Modern vacancy studies use positron annihilation lifetime spectroscopy (PALS). Positrons preferentially trap at vacancy sites, where the reduced electron density increases the annihilation lifetime.

Material (state)	Positron lifetime (ps)	Interpretation
Al (defect-free)	163	Bulk lifetime
Al (monovacancy)	245	Vacancy trapped
Cu (defect-free)	122	Bulk lifetime
Cu (monovacancy)	180	Vacancy trapped
Fe (defect-free)	110	Bulk lifetime
Fe (monovacancy)	175	Vacancy trapped

The lifetime increase of ~50-80% upon trapping at a vacancy provides an unambiguous defect signature.

References

1. R. O. Simmons and R. W. Balluffi, "Measurements of Equilibrium Vacancy Concentrations in Aluminum," Phys. Rev. 117, 52 (1960). DOI: 10.1103/PhysRev.117.52
2. F. Seitz, "Color Centers in Alkali Halide Crystals," Rev. Mod. Phys. 18, 384 (1946). DOI: 10.1103/RevModPhys.18.384

3. E. Mollwo, "Über die Zuordnung von Verfärbungsbanden und Gitterstörungen in Alkalihalogenidkristallen," *Nachr. Ges. Wiss. Göttingen* 1, 97 (1931).
 4. H. F. Ivey, "Spectral Location of the Absorption Due to Color Centers in Alkali Halide Crystals," *Phys. Rev.* 72, 341 (1947). DOI: 10.1103/PhysRev.72.341
 5. R. Krause-Rehberg and H. S. Leipner, *Positron Annihilation in Semiconductors* (Springer, 1999).
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Chapter 21: Dislocations

Theoretical vs Experimental Shear Strength

The theoretical shear strength of a perfect crystal is $\tau_{th} \sim G / (2\pi)$, where G is the shear modulus. Real crystals yield at stresses 100-10,000 times lower due to dislocation motion.

Crystal	G (GPa)	$\tau_{th} = G/2\pi$ (GPa)	τ_{exp} (MPa)	τ_{th} / τ_{exp}
Fe (alpha)	81	12.9	27.5	470
Cu	48	7.6	0.98	7800
Al	26	4.1	0.78	5300
NaCl	13	2.1	0.75	2800
Zn	43	6.8	0.18	38000

This enormous discrepancy, first recognized by Frenkel (1926), motivated the independent proposals of dislocations by Taylor, Orowan, and Polanyi in 1934.

TEM Observation of Dislocations

Transmission electron microscopy provides direct imaging of dislocation lines through diffraction contrast. Taylor and Christian (1967) studied dislocation structures in deformed metals using TEM.

"Electron microscopy reveals that the dislocation density in lightly deformed copper is approximately 10^8 cm^{-2} , increasing to 10^{12} cm^{-2} after heavy cold working." -- Taylor & Christian, Phil. Mag. 15, 893 (1967)

Material (condition)	Dislocation density ρ (cm^{-2})	Method
Annealed Cu single crystal	$10^6 - 10^7$	TEM
Lightly deformed Cu	$10^8 - 10^9$	TEM
Heavily cold-worked Cu	$10^{11} - 10^{12}$	TEM
Si (dislocation-free wafer)	$< 10^0$	X-ray
GaAs (LEC grown)	$10^3 - 10^5$	Etch pit
Stainless steel (annealed)	$10^7 - 10^8$	TEM
Stainless steel (work-hardened)	$10^{10} - 10^{11}$	TEM

Whisker Strength: Approaching Theoretical Limits

Metal whiskers are nearly defect-free single crystals that approach the theoretical shear strength. Brenner (1956) performed systematic tensile tests on metal whiskers.

Material	Whisker diameter (micro-m)	Tensile strength (GPa)	G/ τ ratio	Theoretical max (GPa)
Fe	1.6	13.2	6.1	12.9
Cu	1.25	2.9	16.5	7.6
Ag	4	1.7	17.6	4.8
Sn	15	0.75	25	2.9

Si (nanowire)	0.1	12	5.6	11.2
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Iron whiskers achieve strengths remarkably close to the theoretical limit, confirming that bulk weakness originates from dislocations rather than from intrinsic atomic bonding.

Burgers Vectors and Slip Systems

Crystal structure	Slip plane	Slip direction	Burgers vector $\mathbf{b}$	Number of slip systems
FCC	{111}	$\langle 110 \rangle$	$a/2 \langle 110 \rangle$	12
BCC	{110}	$\langle 111 \rangle$	$a/2 \langle 111 \rangle$	12 (primary)
BCC	{112}, {123}	$\langle 111 \rangle$	$a/2 \langle 111 \rangle$	24, 24
HCP	(0001)	$\langle 11\bar{2}0 \rangle$	$a/3 \langle 11\bar{2}0 \rangle$	3 (basal)
Diamond cubic	{111}	$\langle 110 \rangle$	$a/2 \langle 110 \rangle$	12
NaCl	{110}	$\langle 1\bar{1}0 \rangle$	$a/2 \langle 1\bar{1}0 \rangle$	6

Work Hardening: Stress-Strain Data

The Taylor hardening model predicts the flow stress: $\tau = \alpha G b \sqrt{\rho}$, where $\alpha \sim 0.2\text{--}0.5$.

Material	Stage	Hardening rate $d(\tau)/d(\gamma)$	$\rho \text{ (cm}^{-2}\text{)}$	$\tau \text{ (MPa)}$
Cu	Stage I	$G/40000$	10^7	0.5
Cu	Stage II	$G/300$	10^9	15
Cu	Stage III	decreasing	10^{10}	50
Al	Stage II	$G/200$	10^9	8

Stage I (easy glide) involves single-slip with low hardening. Stage II (linear hardening) corresponds to multi-slip with rapid dislocation multiplication. Stage III (parabolic) reflects dynamic recovery via cross-slip.

Peierls-Nabarro Stress

The lattice friction stress (Peierls stress) controls the intrinsic resistance to dislocation motion:

Material	Peierls stress $\tau_P \text{ (MPa)}$	τ_P / G	Character
Al	0.2	8×10^{-6}	Very mobile
Cu	0.5	1×10^{-5}	Very mobile
Fe (screw)	370	4.6×10^{-3}	High friction
W (screw)	900	5.5×10^{-3}	High friction
Si (300 K)	~ 2000	3×10^{-2}	Covalent
Diamond	$> 10^4$	$> 10^{-2}$	Covalent

BCC metals and covalent crystals show high Peierls stresses for screw dislocations due to their compact, non-planar core structures.

References

1. A. Taylor and B. J. Christian, "Dislocation Arrangements in Deformed Copper," *Phil. Mag.* 15, 893 (1967). DOI:

10.1080/14786436708221636

2. S. S. Brenner, "Tensile Strength of Whiskers," J. Appl. Phys. 27, 1484 (1956). DOI: 10.1063/1.1722294
 3. J. Frenkel, "Zur Theorie der Elastizitatzgrenze und der Festigkeit kristallinischer Korper," Z. Phys. 37, 572 (1926). DOI: 10.1007/BF01397292
 4. G. I. Taylor, "The Mechanism of Plastic Deformation of Crystals," Proc. Roy. Soc. A 145, 362 (1934). DOI: 10.1098/rspa.1934.0106
 5. J. P. Hirth and J. Lothe, Theory of Dislocations, 2nd ed. (Wiley, 1982).
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Chapter 22: Alloys

The Kondo Effect: Resistance Minimum in Dilute Magnetic Alloys

Kondo (1964) explained the anomalous resistance minimum observed in dilute alloys of magnetic impurities in nonmagnetic hosts. The third-order perturbation theory yields a $\ln(T)$ contribution to resistivity that competes with the phonon T^5 term, producing a minimum.

"It is shown that the s-d interaction gives rise to a term proportional to $\log T$ in the resistivity of dilute magnetic alloys." -- J. Kondo, Prog. Theor. Phys. 32, 37 (1964)

The Kondo temperature T_K marks the crossover from weak-coupling (perturbative) to strong-coupling (screened) regimes.

Host-Impurity system	T_K (K)	Impurity spin S	Resistance minimum $T_{\min}$ (K)
Cu:Fe	22	5/2	~10
Cu:Mn	0.01	5/2	~1
Cu:Cr	1-2	3/2	~5
Au:Fe	0.3	5/2	~3
Au:V	300	1	> 300
La:Ce (heavy fermion)	~5	5/2	~5

Systems with high T_K (Au:V) show the resistance minimum at high temperature; low T_K systems (Cu:Mn) exhibit the minimum near 1 K.

Nordheim Rule and Residual Resistivity

The Nordheim rule states that the residual resistivity of a substitutional binary alloy $A_{1-x}B_x$ is proportional to $x(1-x)$:

$$\Delta \rho = C * x(1-x)$$

Alloy system	C (micro-Ohm*cm)	x at maximum	$\rho_{\max}$ (micro-Ohm*cm)
Cu-Au	5.7	0.50	1.43
Cu-Zn	0.30	0.50	0.075
Cu-Ni	1.14	0.50	0.285
Ag-Au	3.6	0.50	0.90
Cu-Ge	4.0	0.50	1.00

The parabolic concentration dependence arises from the Born approximation for scattering from random substitutional disorder.

Order-Disorder Transition in Cu₃Au

Cu₃Au undergoes a first-order order-disorder transition at $T_c = 663$ K. Below T_c , Au atoms occupy corner sites and Cu atoms occupy face-center sites of the cubic unit cell (L1₂ structure).

Property	Ordered ($T < 663$ K)	Disordered ($T > 663$ K)
Structure	L1 ₂ (Pm-3m)	FCC (Fm-3m)

Lattice parameter	3.749 Å	3.753 Å
Long-range order parameter S	1.0 (at 0 K)	0
Electrical resistivity	~3.7 micro-Ohm*cm	~15.5 micro-Ohm*cm
Superlattice reflections (100)	Present	Absent

The resistivity drops by a factor of ~4 upon ordering, because the periodic potential of the ordered alloy drastically reduces electron scattering.

Pb-Sn Eutectic Phase Diagram

The Pb-Sn system is the classic eutectic, long used as solder (now largely replaced by Pb-free alternatives for environmental reasons).

Feature	Composition (wt% Sn)	Temperature (C)
Melting point of Pb	0	327.5
Melting point of Sn	100	231.9
Eutectic point	61.9	183
Max solid sol. Sn in Pb (alpha)	19.2	183
Max solid sol. Pb in Sn (beta)	2.6	183

The eutectic microstructure consists of alternating lamellae of alpha (Pb-rich) and beta (Sn-rich) phases with a characteristic spacing of ~1-5 micro-m depending on cooling rate.

Hume-Rothery Rules: Solid Solution Formation

Hume-Rothery identified empirical conditions for extensive solid solubility:

Rule	Criterion	Example (favorable)	Example (unfavorable)
Atomic size	< 15% difference	Cu-Ni (2.6%)	Cu-Pb (37%)
Electronegativity	Similar values	Cu-Ni	Cu-O
Valence	Same valence preferred	Cu-Ni (both +1/+2)	Cu-Si
Crystal structure	Same structure	Cu-Ni (both FCC)	Cu-Fe (FCC vs BCC)

Cu-Ni satisfies all four rules and forms a complete solid solution at all compositions.

Electron Compounds (Hume-Rothery Phases)

Certain intermetallic phases occur at specific electron-to-atom (e/a) ratios:

Phase type	e/a ratio	Structure	Examples
alpha	< 1.4	FCC solid solution	Cu-Zn (< 35% Zn)
beta	3/2	BCC (CsCl type)	CuZn, Cu ₃ Al, AgCd
gamma	21/13	Complex cubic (52 atoms)	Cu ₅ Zn ₈ , Cu ₉ Al ₄
epsilon	7/4	HCP	CuZn ₃ , AgCd ₃

Heavy Fermion Behavior

In some rare-earth and actinide alloys, the Kondo effect involves the entire lattice of magnetic ions, producing a coherent heavy-fermion state with effective masses $m^* \sim 100\text{-}1000 m_e$.

Compound	γ (mJ/mol $\cdot$ K ²)	m^*/m_e	T _K (K)	Ground state
CeAl ₃	1620	~ 800	5	Heavy Fermi liquid
CeCu ₆	1600	~ 700	5	Heavy Fermi liquid
UPt ₃	450	~ 200	20	Unconventional SC
UBe ₁₃	1100	~ 500	8	Unconventional SC
CeCoIn ₅	290	~ 130	50	d-wave SC

The enormous electronic specific heat coefficient γ reflects the large density of states at the Fermi level from the renormalized f-electron bands.

References

1. J. Kondo, "Resistance Minimum in Dilute Magnetic Alloys," Prog. Theor. Phys. 32, 37 (1964). DOI: 10.1143/PTP.32.37
2. L. Nordheim, "Zur Elektronentheorie der Metalle. II," Ann. Phys. 401, 641 (1931). DOI: 10.1002/andp.19314010503
3. C. S. Barrett, "A Low Temperature Transformation in Lithium," Phys. Rev. 72, 245 (1947). (Cu₃Au data by Cowley, 1950)
4. W. Hume-Rothery et al., "The Freezing Points, Melting Points, and Solid Solubility Limits of the Alloys of Silver and Copper with the Elements of the B Sub-Groups," Phil. Trans. R. Soc. A 233, 1 (1934). DOI: 10.1098/rsta.1934.0014
5. G. R. Stewart, "Heavy-fermion systems," Rev. Mod. Phys. 56, 755 (1984). DOI: 10.1103/RevModPhys.56.755

About the Sources

This book draws on 70 original research papers downloaded from their primary publications. Every data table, every quoted passage, and every numerical value traces back to a specific paper with a DOI. The papers span from W. L. Bragg's 1913 determination of the NaCl crystal structure to 2025 experiments on atomic-scale quantum dots.

Key landmark papers referenced in this book include:

- * Bragg (1913) -- first crystal structure determination (NaCl)
 - * Davisson & Germer (1927) -- electron diffraction proving wave nature of matter
 - * Brockhouse & Stewart (1955) -- first phonon dispersion curve
 - * Dresselhaus, Kip & Kittel (1955) -- cyclotron resonance effective masses
 - * Shull, Strauser & Wollan (1951) -- neutron diffraction of antiferromagnets
 - * Giaever (1960/1974) -- tunneling spectroscopy of superconducting gap
 - * von Klitzing (1980) -- integer quantum Hall effect
 - * Eigler & Schweizer (1990) -- positioning single atoms with STM
 - * Kondo (1964) -- resistance minimum in dilute magnetic alloys
-

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